Financial Conditions, r^*, and the Transmission of Monetary Policy Surprises

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Abstract

Monetary policy is the main tool for stabilizing the economy, but estimates of its effects are troubling. In samples that start in the mid-1990s, standard methods applied to surprises around Federal Reserve announcements give unemployment that falls after a tightening and prices that do not respond. We keep the surprises and change how their effects are traced through the economy. We measure the response of the two-year yield and of a broad financial conditions index (the 10-year yield, equity valuation and the real dollar) in the announcement window, both relative to the neutral rate r^*, and carry it forward with a daily and a monthly model. With the two-year yield alone the surprises give the familiar wrong signs. With the index and r^*, a policy move that raises the two-year yield 1 percentage point raises the unemployment gap 0.34 percentage point after one year and 0.50 after two, and lowers the core price level 0.81 percent after two years; the implied sacrifice ratio is about 0.9 percentage-point years of unemployment per percent of the price level. The results are generally robust to alternative samples, measures of r^*, trends, credit conditions and the weights of the index.

1 Introduction

Monetary policy is the main tool for stabilizing the economy. Its use rests on estimates of how much a change in policy moves unemployment and inflation. Those estimates have become harder to defend. In samples that start after the early 1980s, standard methods often give a surprise tightening small effects on unemployment, or effects of the wrong sign (Ramey, 2016). Its effects on inflation look smaller still. Two concerns follow. One is that policy has become less powerful (Boivin and Giannoni, 2006; Boivin et al., 2010). The other is that inflation responds little to slack: the Phillips curve is flat (Hazell et al., 2022; Del Negro et al., 2020), so lowering inflation requires a large and lasting rise in unemployment. If both were true, policy would be weaker and more costly than standard models suggest. This paper asks whether these findings reflect how policy works or how its effects are measured.

The question matters in practice. From 2022 to 2023 the Federal Reserve raised its policy rate by more than 5 percentage points. Policymakers expected costs: in August 2022 the Chair warned of "some softening of labor market conditions" and called these "the unfortunate costs of reducing inflation" (Powell, 2022). They also described policy as working through broad financial conditions. In November 2023 the Chair noted that "Financial conditions have tightened significantly in recent months, driven by higher longer-term bond yields, among other factors" (Powell, 2023). Yet the standard tools for measuring the effects of policy leave those conditions out.

Interest-rate surprises around Federal Reserve announcements were meant to settle the measurement problem. The FOMC makes its decision before the announcement, so market moves in a narrow window around the announcement, at most about 100 minutes long, cannot have caused

the decision. The change in interest-rate futures and Treasury yields in that window is then news about policy, or at least about the part of policy that moves interest rates (Kuttner, 2001; Gürkaynak et al., 2005). These surprises measure the effects of policy on asset prices precisely (Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018; Swanson, 2021). But when they are used to measure effects on unemployment and inflation, the old puzzles often return. Recent work offers several explanations, each with a fix, and the fixes complement one another.

- *Information.* An announcement reveals the Fed's view of the economy as well as its policy, so a surprise tightening can signal good news (Romer and Romer, 2000; Nakamura and Steinsson, 2018). Separating the two gives larger price effects (Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021).
- *Predictable surprises.* Part of each surprise can be predicted from earlier public data. Removing it, and adding surprises from the Chair's speeches, gives larger and more precise effects (Bauer and Swanson, 2023a,b).
- *Timing.* Markets react within minutes, but unemployment and inflation are monthly averages. Measuring inflation daily removes most of the wrong-signed response (Jacobson et al., 2026).
- *Credit and risk appetite.* Credit costs carry part of the effect (Gertler and Karadi, 2015; Caldara and Herbst, 2019). Announcements also move stock prices and the dollar much more than interest-rate surprises can explain. Treated as a separate shock, this news has large effects (Kroencke et al., 2021; Bauer et al., 2023; Boehm and Kroner, 2024; Bauer et al., 2026; Boehm et al., 2026).

Most of this work, however, traces the surprise forward through a single interest rate, measured in nominal terms. That leaves out two basic parts of how monetary policy works.

First, policy affects spending through financial conditions as a whole: the cost of long-term credit, the value of wealth and the exchange rate (Boivin et al., 2010). A surprise tightening moves long-term yields, stock prices and the dollar, not only the two-year yield. A model that omits these variables traces the wrong experiment. The financial conditions index (FCI) used below combines the 10-year yield, stock valuations and the dollar. It is scaled so that, in the announcement windows, it rises about 1 unit per percentage point rise in the two-year yield. In an ordinary month an unexpected rise of 1 percentage point in the two-year yield comes with a rise in the FCI of only 0.36 to 0.51 unit at the real and r^* steps. A model with the two-year yield alone therefore traces the typical move in that yield, which comes with a much smaller change in financial conditions than a policy surprise brings.

Second, what matters for spending and inflation is the level of real interest rates compared with the neutral rate r^*, the rate that neither speeds up nor slows down the economy. The neutral rate fell a lot from the 1990s to the 2010s (Laubach and Williams, 2003; Holston et al., 2017, 2023; Hamilton et al., 2016; Kiley, 2020b), and it shaped the long-run trend in yields (Bauer and Rudebusch, 2020). Since the mid-1990s, interest rates have trended down along with r^*, while activity has trended up. A model that links the level of rates to activity then partly picks up these opposite trends, and finds that higher rates go with stronger activity. Activity responds to changes in financial conditions, but inflation depends on how long real rates stay above or below r^*. So measuring rates relative to r^* should matter most for prices.

Our contribution is to hold the surprise fixed and change only the variables that carry it to the economy. Broad financial conditions, measured relative to r^*, replace a single nominal rate. The shock is the policy-news surprise of Nakamura and Steinsson (2018), measured in narrow windows around 252 FOMC announcements from 1996 to 2026. We measure its effect on the two-year yield and on the FCI within these windows, and assume that unemployment and inflation do not

move within them. A daily model then traces how the two-year yield and the FCI evolve after each announcement. Monthly unemployment and core inflation respond to the monthly averages of these daily series, as in a standard vector autoregression (VAR). Like Jacobson et al. (2026), we take into account when each announcement falls within its month. The FCI and the order of steps (nominal, real, then relative to r^*) come from a companion note on financial conditions and were fixed before this analysis (FCI note, 2026). "Significant" below means that the 90 percent interval excludes zero.

We begin with the standard methods on our own data, for the sample that starts in 1996. In a recursive VAR and in a proxy VAR that uses the same surprises as instrument, unemployment falls after a tightening, by 0.95 and 0.91 percentage point after 12 months per percentage point on the two-year yield, and the price level does not respond significantly. The model with the nominal two-year yield alone gives the same pattern. Adding the FCI removes the fall in unemployment at every step, and measuring both the yield and the FCI relative to r^* gives the baseline: unemployment rises 0.34 percentage point after 12 months and 0.50 after 24, and the core price level falls 0.81 percent after 24 months and 1.11 after 36, all significant at 24 months or later.

The results also bear on the inflation-unemployment tradeoff. We measure it by the "Phillips multiplier" of Barnichon and Mesters (2021): the fall in the price level per percentage-point-year of higher unemployment. In the baseline the multiplier is -1.07 after two years, with a 90 percent set of $[-5.90, -0.41]$ that excludes zero, a sacrifice ratio of about 0.9 percentage-point years of unemployment per percent of the price level. With the nominal or real FCI, or with the yield alone, the data cannot pin the tradeoff down.

The results are generally robust to alternative samples, measures of r^*, trends, credit conditions and the weights of the FCI (Section 4). A monthly proxy VAR that adds the FCI is a noisy way to estimate these effects: in a simulation in which our model is the true process, its estimates are centred on the truth but spread widely.

Section 2 sets out the approach, Section 3 gives the results, and Section 4 the robustness checks. Section 5.1 draws conclusions for policy.

1.1 Connection to the literature

Identifying the effects of policy. Our paper is closest in spirit to Jacobson et al. (2026). Like them, we hold the surprise and the method fixed and change one measurement. They change how often the outcome is measured; we change how the channel from policy to the economy is measured. They study inflation only, while our main result is for activity. Both changes remove wrong-signed responses, and both point to a mismatch between what the economist measures and what drives decisions. Bauer and Swanson (2023a) and Jarociński and Karadi (2020) improve the surprise itself. In our design a surprise enters only through its immediate effect on the two-year yield and the FCI, and through its timing. Their fixes and ours can in principle be combined. Our approach also changes which assumptions the estimates rest on. It depends less on how strongly the surprise moves rates over a month, and less on its link to earlier news. It relies instead on the assumption that the economy responds to a given change in financial conditions in the same way, whatever caused it.

The use of futures-based surprises in monthly models goes back to Faust et al. (2004). The choice of surprise series can matter (Bu et al., 2021; Brennan et al., 2025), and so can the number of lags (Coibion, 2012). Related work on timing and mixed frequencies includes Kilian (2024), Chudik and Georgiadis (2022), Francis et al. (2011), Ghysels (2016) and Gazzani and Vicondoa (2020).

Financial channels and risk appetite. Gertler and Karadi (2015) and Caldara and Herbst (2019) are closest in emphasizing financial channels. We use a broad index and measure it relative

to r^*, but leave credit spreads out of the baseline. Recent studies of risk appetite and the "non-yield" shock, the announcement news that moves stocks and the dollar beyond what interest rates explain, differ from ours in where they place risk premia (Bauer et al., 2026; Boehm and Kroner, 2024; Boehm et al., 2026). They treat policy-driven moves in risk premia as a second shock; we treat them as part of financial conditions. The channels are the same in both views: stock prices, the dollar and credit costs. Our estimates keep only the part of the stock and dollar moves that comes with interest-rate news, so they measure the effects of that dimension of policy, not of all policy news. The response of stock prices to policy-driven moves in long rates has also changed over time (Kiley, 2014). Section 2.2.4 shows how large the rest is. Caballero et al. (2026) propose a neutral level of the FCI itself, FCI*, because r^* must otherwise absorb asset-price movements that the index already captures. We keep r^* for the rates inside the two-year yield and the index; using gaps to FCI* is a natural extension.

The power of policy and the Phillips curve. A long literature finds that the effects of policy shocks became smaller after the early 1980s. It attributes the change to a more systematic response of policy and to better-anchored expectations (Boivin and Giannoni, 2006; Boivin et al., 2010; Primiceri, 2005). Others argue that standard identification fails after the late 1980s because policy became more forward-looking (Barakchian and Crowe, 2013).

On the Phillips curve, recent work finds a small slope (Hazell et al., 2022; Del Negro et al., 2020; Ball and Mazumder, 2019; Fitzgerald et al., 2024). It also stresses that the slope is hard to identify when policy offsets demand shocks (McLeay and Tenreyro, 2020), and that conventional methods tend to understate it (Barnichon and Mesters, 2020). Policy shocks offer one way to identify it. Barnichon and Mesters (2021) measure the tradeoff by the ratio of cumulative inflation to cumulative unemployment after such shocks. In that metric they find a much flatter tradeoff after 1990 than before. Responses to policy shocks also map directly into the outcomes of alternative policies (McKay and Wolf, 2023). We add evidence of this kind and show that the measured tradeoff depends on how the stance of policy is measured. Our price result relies on 2020–22, when inflation surged and then began to fall. Others read that episode as evidence of a steeper, nonlinear Phillips curve (Benigno and Eggertsson, 2023), or as the unwinding of supply shocks (Bernanke and Blanchard, 2025); we return to this in Section 4.1.

2 Empirical approach

The model has three pieces. A surprise moves the two-year yield and the FCI by amounts measured in the announcement window: the vector b in equation (13). A daily VAR, equation (14), carries those two series forward. Their monthly averages then enter a monthly VAR for unemployment and inflation, equation (15). Section 2.1 defines the FCI, Section 2.3 sets out the timing assumptions at each frequency, and Section 2.4 gives the model. Appendix B gives the daily construction of the series, the algorithm for tracing responses and the bootstrap. Appendix C lists the data.

2.1 Measuring financial conditions

The FCI is the market index of FCI note (2026): a weighted sum of the 10-year Treasury yield, stock valuations and the real broad dollar. Indexes of this kind are weighted sums of financial variables (Hatzius et al., 2010). They differ mainly in where the weights come from. FCI-G takes them from large macroeconomic models (Ajello et al., 2023). Ours come from the event-study regressions long used to measure how asset prices respond to policy (Kuttner, 2001; Bernanke and Kuttner, 2005; Hanson and Stein, 2015; Nakamura and Steinsson, 2018).

Policy news. Let ΔX_a be the 5×1 vector of changes in the current- and next-meeting federal funds futures and the second to fourth eurodollar futures in the window around announcement a. Following Nakamura and Steinsson (2018), let p_a be the first principal component of these changes, signed so that it moves with the fourth eurodollar future. Policy news is p_a rescaled so that one unit moves the two-year yield y^2 by 1 percentage point:

$$s_a = \gamma\, p_a, \qquad \gamma = \frac{\sum_a p_a\, \Delta y_a^2}{\sum_a p_a^2}. \tag{1}$$

Weights. Let ΔA_a be the vector of changes in the 10-year yield, the S&P 500 and the DXY dollar index in the announcement window. The companion note treats these changes as

$$\Delta A_a = \psi\, s_a + \eta_a, \tag{2}$$

where ψ holds the policy responses and η_a is other news, with covariance matrix Σ. It estimates ψ by regressing each element of ΔA_a on s_a and a constant, and Σ from the residuals, over 253 announcements from January 1996 to September 2026. The weights are those that recover the policy component most precisely (one-factor generalized least squares):

$$w = \frac{\Sigma^{-1}\psi}{\psi' \Sigma^{-1} \psi}. \tag{3}$$

Because $w'\psi = 1$ and s_a is scaled to the two-year yield, a policy move that raises the two-year yield 1 percentage point raises the index by 1 unit. The weights are 0.61 on the 10-year yield, -0.051 per percent on stock valuations and 0.065 per percent on the dollar. Each of the three accounts for about a third of the index's response to policy.

Three versions. In month t, the index in version v is

$$F_t^v = w_B\, B_t^v + w_Q\, Q_t^v + w_D\, D_t, \tag{4}$$

measured as a deviation from its sample mean. Here D_t is 100 times the log of the real broad dollar in every version. With y_t^{10} the 10-year yield, π_t^{10} 10-year expected inflation, and $(E/P)_t$ the inverse of the cyclically adjusted price-earnings ratio (CAPE), the other two terms are:

$$\text{Nominal:} \quad B_t = y_t^{10}, \qquad Q_t = 100 \ln \text{CAPE}_t, \tag{5}$$

$$\text{Real:} \quad B_t = y_t^{10} - \pi_t^{10}, \qquad Q_t = 100 \ln \text{CAPE}_t, \tag{6}$$

$$r^*\text{-gap (baseline):} \quad B_t = y_t^{10} - \pi_t^{10} - r_t^*, \qquad Q_t = -\frac{100}{\overline{E/P}} \left[(E/P)_t - r_t^* \right]. \tag{7}$$

In the r^*-gap version, stocks are measured by the gap between their earnings yield and r^*. The term $100/\overline{E/P}$, with $\overline{E/P}$ the sample average, converts this gap to the units of the log CAPE, so that the weight on stocks is the same in every version. CAPE is a real valuation measure, so stocks do not change between the nominal and real versions. For r^* we use the New York Fed's current Laubach–Williams estimates (Holston et al., 2023). These "one-sided" estimates use only data up to each date, with the model's parameters estimated on the full sample.

The two-year yield. The two-year yield is adjusted in the same steps. With y_t^2 the nominal yield and π_t^2 two-year expected inflation (from the Cleveland Fed; Haubrich et al., 2012),

$$R_t^{\text{nominal}} = y_t^2, \qquad R_t^{\text{real}} = y_t^2 - \pi_t^2, \qquad R_t^{r^*\text{-gap}} = y_t^2 - \pi_t^2 - r_t^*. \tag{8}$$

Expected inflation and r^* are monthly or quarterly and move slowly. The daily series add their interpolated monthly values to daily market prices, so day-to-day moves come from markets (Appendix B).

2.2 Why financial conditions and r^*

This section sets out the reasoning behind the design. We first describe how monetary policy is thought to reach spending and prices. We then show why a standard VAR can miss this route, and why that could produce the puzzles. Finally, we connect the argument to work on financial conditions and economic activity.

2.2.1 The standard transmission mechanism

Monetary policy sets the short-term interest rate today and, through its communications, shapes what people expect that rate to be in the future. Few households and businesses borrow at the policy rate. They face the rates on consumer credit, mortgages and business debt, which arbitrage links to current and expected short-term rates. The same rise in rates raises the return investors require on stocks. For businesses, a higher cost of capital raises the user cost of new equipment and buildings, and investment falls (Jorgenson, 1963; Hall and Jorgenson, 1967). The value of existing capital relative to its replacement cost falls as well (Tobin, 1969). For households, lower stock prices lower wealth, and consumption depends on wealth as well as on current income (Ando and Modigliani, 1963; Friedman, 1957). Higher rates also strengthen the dollar, which shifts spending toward imports and reduces exports (Fleming, 1962; Mundell, 1963).

Interest rates of many maturities, equity prices and the dollar therefore form the core of the transmission mechanism (Boivin et al., 2010). These links run from the early work cited above to modern models of the business cycle, in closed and open economies (Christiano et al., 2005; Smets and Wouters, 2007; Galí and Monacelli, 2005). Models with heterogeneous households give more weight to the indirect effects that work through income (Kaplan et al., 2018; Auclert, 2019). But in all of these models the short-term rate matters mainly because of what it does to the prices and rates that people actually face. And they act with a lag (Friedman, 1961).

What matters in these models is not the nominal rate but the real rate relative to the neutral rate r^*. The same holds for the other parts of financial conditions: a given yield or stock valuation is tighter when r^* is low than when it is high. The distinction matters most for inflation. Spending responds to financial conditions within months. Inflation responds to the slack that builds up over time, so it depends on how long, and by how much, conditions stay tight relative to r^*. A measure of conditions that ignores a falling r^* will overstate how tight policy has been, and by more the longer the period. That error is largest for the cumulative effects that drive inflation.

2.2.2 What standard VARs leave out

The workhorse evidence on monetary policy comes from VARs with a few macroeconomic variables and one interest rate (Bernanke and Blinder, 1992; Christiano et al., 1999). Some studies add a credit spread (Gertler and Karadi, 2015; Caldara and Herbst, 2019), and factor models bring in many more series (Bernanke et al., 2005). But long-term yields, equity valuations and the dollar are usually absent from the small models used to measure the effects of policy, although some recent work adds the dollar and stock-market volatility (Boehm et al., 2026). These are the variables that, on the mechanism above, carry those effects. Concern about missing variables is not new: Sims (1992) argued that the price puzzle arises because the Federal Reserve responds to information that small VARs do not contain (see also Castelnuovo and Surico, 2010). Our concern is different. Here the missing variables are the channel itself.

A simple model shows what this omission can do. Let U_t be the unemployment gap, i_t the

policy rate and f_t financial conditions, with a higher f_t meaning tighter conditions:

$$U_t = \rho_U \, U_{t-1} + \kappa \, f_{t-1} + \varepsilon_t^U, \tag{9}$$

$$f_t = \rho_f \, f_{t-1} + \lambda \, i_t + \varepsilon_t^f. \tag{10}$$

Unemployment responds to financial conditions and not directly to the policy rate. Financial conditions respond to the policy rate and to other forces, ε_t^f, such as changes in risk appetite. Some of these other forces may themselves be policy news that does not move interest rates (Boehm and Kroner, 2024; Bauer et al., 2026). Suppose the economist observes only U_t and i_t. To remove f_t, lag equation (9) one month, multiply it by ρ_f, subtract it from equation (9), and use equation (10). This gives

$$U_t = (\rho_U + \rho_f) \, U_{t-1} - \rho_U \rho_f \, U_{t-2} + \kappa\lambda \, i_{t-1} + \underbrace{\varepsilon_t^U - \rho_f \, \varepsilon_{t-1}^U + \kappa \, \varepsilon_{t-1}^f}_{\text{error}}. \tag{11}$$

The policy rate has no direct effect in this model. It appears in equation (11) because it stands in for financial conditions. The error is a moving average that includes past financial shocks, so a VAR in U_t and i_t with few lags cannot capture it. The VAR is misspecified. With the lags and samples available in practice, this can make the effects of monetary policy harder to measure.

A second problem matters more for the size and sign of the response: the omission changes the experiment being traced. In the announcement windows, the FCI used below rises about 1 unit per percentage point rise in the two-year yield. In an ordinary month, an unexpected rise of 1 percentage point in the two-year yield, measured by the residual of a monthly VAR in unemployment, inflation, the yield and the FCI, comes with a rise in the FCI of only 0.36 to 0.51 unit at the real and r^* steps (0.38 at the nominal step). A VAR without financial conditions traces the typical move in the two-year yield, which comes with a much smaller change in financial conditions than a policy surprise does.

The two-equation model illustrates how an omitted variable distorts the estimated response.

High-frequency surprises help identify the shock, but they do not by themselves fix how it is carried forward. Our design therefore does two things. It includes financial conditions in the system, so that they carry the effects of the surprise forward, as in equation (9). And it measures the immediate effect of the surprise on the two-year yield and on financial conditions directly in the announcement window, where policy news is the main force at work. The measurement of f_t also matters. If activity responds to financial conditions relative to r^*, a measure that ignores r^* differs from the true f_t by a slow-moving error, and that error enters equation (9).

2.2.3 Financial conditions and real activity

A separate literature asks how financial conditions are related to economic activity. Asset prices help forecast output and inflation, but which ones help changes over time (Stock and Watson, 2003). Credit spreads, and especially their excess bond premium component, are strong predictors of activity (Gilchrist and Zakrajšek, 2012). Broad indexes summarize many such series and are used to assess the outlook (Hatzius et al., 2010; Brave and Butters, 2011). Financial conditions are also linked to the risks around growth, not only to its average (Adrian et al., 2019), and their relationship with unemployment may be nonlinear (Kiley, 2020a). Federal Reserve staff now publish a measure of the effect of financial conditions on growth, with weights from large macroeconomic models (Ajello et al., 2023).

This work has a limit that the event-study literature does not. A correlation between financial conditions and later activity need not reflect a causal effect, because financial conditions also respond to the economy. Hatzius et al. (2010) address part of this problem by removing the effects

of past growth and inflation from their indicators. The companion note uses the timing in a VAR for the same purpose (FCI note, 2026). The two literatures have complementary strengths. Work on high-frequency surprises identifies policy shocks cleanly but carries them through a small set of variables. Work on financial conditions studies a broad set of variables but does not isolate policy shocks. We bring them together: the weights in our index come from how its components respond to policy news, and the surprise is carried forward through the estimated relationship between that index and the economy.

2.2.4 Policy news beyond interest rates, and risk appetite

Recent work finds that announcements move stock prices and the dollar much more than interest-rate surprises can explain (Boehm and Kroner, 2024; Bauer et al., 2026). Our windows show the same pattern. The policy-news surprise explains 80 percent of the variance of the move in the two-year yield, but only 32 percent for the 10-year yield, 31 percent for the S&P 500, 34 percent for the dollar and 48 percent for the FCI. The part of the FCI's move that the surprise does not explain is therefore large.

There are two readings of this unexplained part. On one, it is mostly news unrelated to policy that happens to arrive in the window, and our estimates capture the effects of policy. On the other, much of it is policy news that does not move interest rates, as Boehm and Kroner (2024) and Bauer et al. (2026) argue. Then our estimates are the effects of one dimension of policy news, and the effects of all policy news through financial conditions could be larger. Both readings agree on the channel: policy news reaches the economy through broad financial conditions.

[TO DO: The earlier draft also reported (i) the window ratio by subperiod and by VIX/EBP state, (ii) the effect of a one-standard-deviation move in the unexplained part of the FCI, (iii) Baa/EBP credit, FCI-components and commodity-price robustness. None of these is in the clean package; (iii) credit moves to the EBP robustness subsection (Section 4); regenerate (i) and (ii) in baseline/ only if the paper needs them.]

2.3 Timing assumptions and identification

The shock is the Nakamura–Steinsson policy-news surprise, s_a in equation (1), from the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). We use the 252 FOMC announcements from January 1996 to July 2026 with all the window measures available (Appendix B). The window runs from 10 minutes before the statement to 60 minutes after the press conference starts, or 30 minutes around the statement when there is no press conference. Identification comes from timing, at three frequencies. Table 1 summarizes the assumptions.

Table 1: Timing assumptions

Frequency	What can move	Assumption	Equation
Window (minutes)	Two-year yield and FCI, by the amounts measured in the window	News about the path of interest rates is the main systematic force in the window, and the only one correlated with the surprise. Unemployment and inflation do not move.	(13)
Daily	Two-year yield and FCI, each day after the announcement	They depend on their own recent history and on the unemployment and inflation of this month and last month.	(14)
Monthly	Unemployment and inflation, from the month after the announcement	Within a month, unemployment and inflation are set first. They respond to financial variables only with a lag of at least one month.	(15)

The window. The window move in the FCI is computed with the weights in equation (3):

$$\Delta F_a = w_B \, \Delta y_a^{10} + w_Q \, \Delta \mathrm{SP}_a + w_D \, \Delta \mathrm{DXY}_a. \tag{12}$$

These are the measures available within the day, and the ones used to estimate the weights. Percent changes in the S&P 500 equal percent changes in CAPE within a window, because earnings are fixed. We express all effects per percentage point on the two-year yield in the window. The effect on the FCI is the ratio of two regression slopes on the surprise, an instrumental-variables estimate with s_a as the instrument:

$$b = \begin{pmatrix} 1 \\ b_F \end{pmatrix}, \qquad b_F = \frac{\sum_a (\Delta F_a - \overline{\Delta F})\,(s_a - \bar{s})}{\sum_a (\Delta y_a^2 - \overline{\Delta y^2})\,(s_a - \bar{s})} = 0.99. \tag{13}$$

This is close to one because the weights are normalized that way. It is not exactly one because this regression includes a constant, which the scaling in equation (1) does not, and because it uses one fewer announcement. Expected inflation and r^* move slowly, so we take the window move to be the same in every version.

The day. On the announcement day, the two-year yield and the FCI shift by the vector b. We use the window move rather than the change over the whole day, because the full-day change contains other news. On later days, the two series move as equation (14) implies. They affect each other only through the daily lags, so no ordering between them is needed within the day. The window fixes their joint move.

The month. Within each month, unemployment and inflation are set first, and markets then respond to them. So in the month of the announcement, unemployment and inflation have not moved. They respond from the next month on, through the monthly average of the daily financial series. That average depends on when in the month the announcement falls: an announcement late in a month affects few days of that month's average. This is how the model takes the timing of announcements into account, as Jacobson et al. (2026) recommend. The ordering is the familiar recursive one, in which slow-moving macroeconomic variables come before financial variables (Bernanke and Blinder, 1992; Christiano et al., 1999). It also means that the daily equations include the current month's unemployment and inflation, even though those figures are published later. We treat this as a modeling convention: the monthly values summarize conditions that markets observe as the month unfolds.

The key assumptions. Two assumptions carry the results. The first is the usual one in a VAR. Once the shock has been identified, the economy responds to a change in the financial variables in the same way whatever caused it. A policy shock identified by a Cholesky ordering or an external instrument is traced through the same estimated dynamics as every other shock, and we do the same. This has a familiar cost. Stock prices partly anticipate future activity, so a move that policy news pushes through stocks need not act on the economy like a move with another cause. This is the same problem as the information effect in any VAR with a high-frequency shock; as in the companion note, we acknowledge it and set it aside.

The second assumption is specific to this paper and has two parts. A standard VAR estimates the effect of the shock on every variable in the system. Here, the surprise reaches unemployment and prices only through the two-year yield and the FCI, and has no direct effect on them in the window. Anything else it moves, such as credit spreads or expectations, is left out. So are the moves in stock prices and the dollar in the window that are unrelated to s_a; the surprise explains only about a third of those moves (Section 2.2.4). The non-yield shock of Boehm and Kroner (2024) is uncorrelated with policy-news surprises, so such moves do not bias b_F. But the weights in equation (3) treat them as noise, and part of them may be policy news (Boehm and Kroner, 2024; Bauer et al., 2026). In addition, the monthly equations see the FCI as a single index. Its weights show how the three components respond to policy news, not how the economy responds to each. So we assume that the three affect unemployment and prices in proportion to those weights. Section 2.2.4 discusses the first part, and Section 4 relaxes the second by entering the three components of the FCI separately.

2.4 A mixed-frequency VAR

Structure. The model has a daily block and a monthly block. Let $x_\tau = (R_\tau, F_\tau)'$ hold the two-year yield and the FCI on day τ, in the same version. Let $m_t = (U_t, \pi_t)'$ hold the two monthly variables. The unemployment gap U_t is the unemployment rate minus the CBO estimate of its natural rate. The CBO estimate does not respond to the surprise, so the response of the gap equals that of the unemployment rate. Inflation is $\pi_t = 1200\,\Delta \ln P_t$, where P_t is the price index for personal consumption expenditures excluding food and energy (core PCE).

The daily block is a VAR in the style of the heterogeneous autoregression of Corsi (2009):

$$x_\tau = \alpha + \sum_{k\in\{1,5,22,66\}} \Phi_k\, \bar{x}^{(k)}_{\tau-1} + \Gamma_0\, m_{t(\tau)} + \Gamma_1\, m_{t(\tau)-1} + \xi_\tau, \tag{14}$$

where $\bar{x}^{(k)}_{\tau-1}$ is the average of x over the k days through $\tau-1$, and $t(\tau)$ is the month of day τ. Averages over about a day, a week, a month and a quarter capture persistence with few coefficients. When the FCI is left out, x_τ is the two-year yield alone.

The monthly block is

$$m_t = c + \sum_{l=1}^{L} A_l\, m_{t-l} + \sum_{l=1}^{L} G_l\, \bar{x}_{t-l} + \theta_1\, t + \theta_2\, t^2 + \zeta_t, \qquad \bar{x}_t = \frac{1}{n_t}\sum_{\tau\in t} x_\tau, \tag{15}$$

with n_t the number of weekdays in month t. The model is a mixed-frequency VAR (Ghysels, 2016). In the language of mixed-data sampling, the weights are equal across days within a month and unrestricted across months (Ghysels et al., 2007; Foroni et al., 2015). The equation contains no current-month term in $\bar{x}_t$, which is the timing assumption of Table 1.

Relation to a monthly VAR and a proxy VAR. Suppose the daily block were replaced by a monthly equation for $\bar{x}_t$, and every announcement occurred on the first day of the month. With

$Z_t = (m_t', \bar{x}_t')'$, the model would then be a standard monthly VAR with a recursive ordering, the macroeconomic variables first, and impact vector $(0, 0, b')'$. The daily block adds two things: the position of each announcement in its month, and fast dynamics of the financial variables, estimated on many more observations. A proxy VAR (Gertler and Karadi, 2015; Mertens and Ravn, 2013; Stock and Watson, 2018) instead estimates $Z_t = \sum_l C_l Z_{t-l} + \iota_t$ and uses as an instrument z_t the surprises summed over the month. The effect on the FCI per unit effect on the two-year yield is then

$$\frac{\mathrm{Cov}(\iota_{F,t}, z_t)}{\mathrm{Cov}(\iota_{R,t}, z_t)}. \tag{16}$$

The window ratio b_F in equation (13) has the same form, with three differences. It replaces monthly VAR residuals with changes measured inside the window, which do not depend on how strongly the surprise predicts monthly residuals. It sets the entries for unemployment and inflation to zero rather than estimating them. And it dates each shock to its day rather than summing shocks over a month. Section 3.5 estimates both monthly benchmarks.

The time trend. The baseline monthly equations contain a constant and no time trend. Over 1996–2026, unemployment, inflation and interest rates contain slow movements that a short monthly model may not capture: rates drifted down along with r^*, inflation was low in the 2010s and rose after 2020, and the natural rate of unemployment may have shifted in ways the CBO estimate misses. Section 4 reports a cubic spline trend through the business-cycle peaks, with the real measures of the yield and the FCI.

Estimation and inference. We estimate each block by ordinary least squares, on the same sample: January 1996, the month of the first announcement with all window measures, to September 2026 (369 months, and the weekdays of those months for the daily block). The lagged daily averages and the monthly lags may use earlier observations. 2020 is kept in both blocks; Section 4 drops it. The baseline has $L = 12$ monthly lags, fixed in advance; the appendix shows 3 and 6. Responses are computed recursively, month by month and day by day, as deviations from a path with no surprise, and averaged over the positions of the announcements within their months (Appendix B). The response of the price level after h months is $\sum_{j=0}^{h} \tilde{\pi}_j/12$. A bootstrap with 200 draws resamples the announcements, which gives new values of b and of the announcement positions, and the residuals of the monthly block. We report 90 percent intervals, as recent studies in this area do (Bauer and Swanson, 2023a; Caldara and Herbst, 2019; Miranda-Agrippino and Ricco, 2021; Ramey, 2016), and 68 percent intervals, which Jarociński and Karadi (2020) report alongside them. The FCI weights and r^* are held fixed. In large samples a VAR of this kind estimates the same effects as local projections (Plagborg-Møller and Wolf, 2021).

3 Baseline results

3.1 Standard methods after the early 1980s

Standard identification schemes have not delivered conventional effects of monetary policy in samples that start after the early 1980s. Ramey (2016) makes the point with the specification of Christiano et al. (1999): estimated on 1965–1995 it gives the classic responses, but on 1983–2007 "the results change dramatically and imply that increases in the federal funds rate lower the unemployment rate"; a version of the Romer–Romer hybrid VAR gives, on the same later sample, contractionary shocks that "appear to be expansionary", as Barakchian and Crowe (2013) found for 1988–2007; and a proxy SVAR with the Romer–Romer shock as instrument, estimated on 1969–2007, gives a "pronounced, statistically significant price puzzle" (her Figures 1 and 2). Her reading is that, because policy has been conducted well, true monetary shocks are rare, and it is hard

to extract shocks that are not contaminated by the central bank's foresight. Caldara and Herbst (2019) find, for 1994–2007 with a proxy SVAR that uses high-frequency surprises, that without a credit spread in the system the response of industrial production is weak and uncertain (about −0.2 percent at the posterior mean, with a credible set that includes positive values), against about −0.75 percent when the excess bond premium is added.

We start from the same two standard methods, estimated on our own data and units, so that they can be compared directly with the model below. Both use the monthly unemployment gap, core PCE inflation and the end-of-month two-year Treasury yield, with 12 lags and no trend. The first is the recursive (Cholesky) VAR of Christiano et al. (1999), with the policy rate ordered last, so that unemployment and inflation do not respond to it within the month. The second is a proxy VAR (Stock and Watson, 2018; Mertens and Ravn, 2013; Gertler and Karadi, 2015) in which the instrument is the Nakamura–Steinsson policy news (Nakamura and Steinsson, 2018) summed over the announcements of the month, the same 252 announcements used throughout. All responses are to a 1 percentage point rise in the policy rate. Figure 1 and Table 2 show the results.

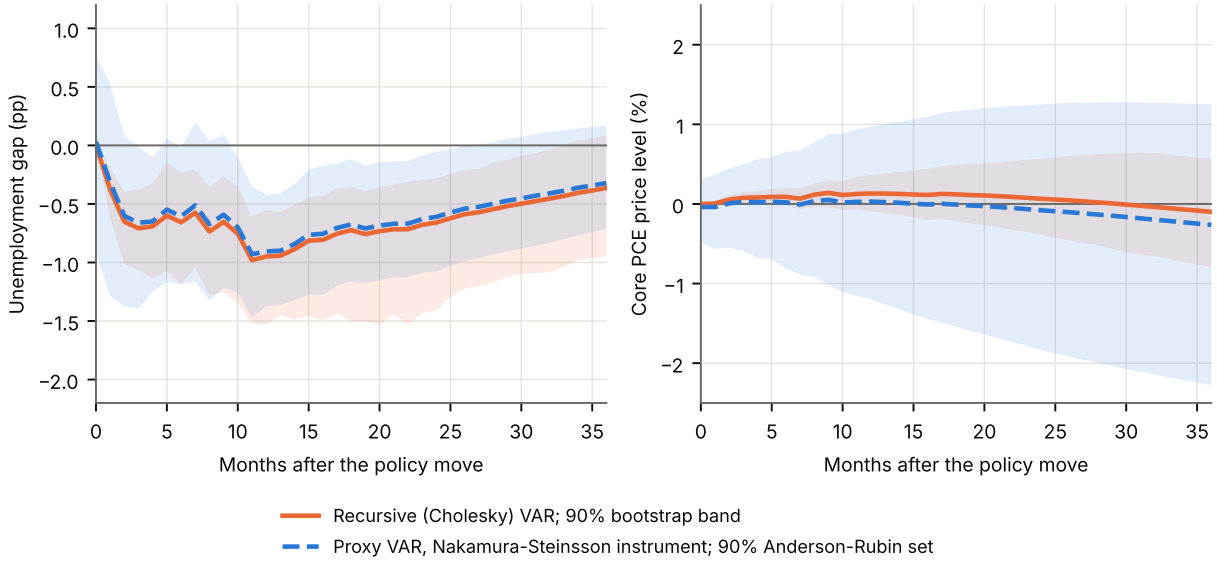


Figure 1: Responses to a 1 percentage point rise in the two-year yield, 1996–2026, from a recursive (Cholesky) VAR and a proxy VAR with the Nakamura–Steinsson instrument. The monthly unemployment gap, core PCE inflation and the two-year yield; 12 lags. Bands: 90 percent (bootstrap for the Cholesky VAR, Anderson–Rubin sets for the proxy VAR, truncated at ±4).

Table 2: Standard methods after the early 1980s: responses to a 1 percentage point rise in the policy rate

			Unemployment gap (pp)			Price level (%)		
Method	Sample	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Cholesky, 2-year yield	1996–2026	369	−0.95*	−0.66*	−0.36	0.13	0.07	−0.10
			[−1.53, −0.55]	[−1.41, −0.28]	[−0.94, 0.09]	[−0.09, 0.36]	[−0.39, 0.57]	[−0.79, 0.58]
Cholesky, 2-year yield	1984–2026	513	−0.61*	−0.36*	−0.16	0.11	0.15	0.13
			[−0.98, −0.33]	[−0.78, −0.10]	[−0.63, 0.06]	[−0.02, 0.29]	[−0.10, 0.47]	[−0.26, 0.57]
Cholesky, federal funds rate	1984–2007	288	−0.04	−0.02	0.00	−0.01	−0.01	0.00
			[−0.07, 0.00]	[−0.06, 0.02]	[−0.05, 0.03]	[−0.06, 0.03]	[−0.09, 0.07]	[−0.13, 0.10]
Proxy VAR, 2-year yield	1996–2026	369	−0.91*	−0.61*	−0.32	0.03	−0.08	−0.26
			[−1.37, −0.42]	[−1.08, −0.05]	[−0.71, 0.17]	[−1.20, 0.97]	[−1.83, 1.25]	[−2.27, 1.25]

Monthly data, 12 lags, constant, no trend; unemployment gap, core PCE inflation and the end-of-month policy rate (the 2-year Treasury yield, or the effective federal funds rate where stated), policy rate ordered last in the Cholesky VARs; price level = cumulated inflation/12. Proxy VAR: Nakamura–Steinsson policy news summed within the month as instrument; first-stage $F = 4.5$. Unit: +1 pp on the policy rate on impact. Brackets: 90% intervals (residual bootstrap for the Cholesky VARs; exact Anderson–Rubin sets for the proxy VAR). * = excludes zero. Specification: baseline/lib/litstyle.py.

In both, unemployment *falls* after a tightening: by 0.95 percentage point after 12 months in the Cholesky VAR (significant) and by 0.91 in the proxy VAR (significant). The responses are significantly negative at 24 months as well, and the Cholesky VAR gives the same answer when the sample starts in 1984 (−0.61 after 12 months). The price level does not respond significantly in any of them (0.13 and 0.03 after 12 months). With the federal funds rate on a 1984–2007 sample, close to the 1983–2007 sample of Ramey (2016), the Cholesky VAR gives the same sign in smaller numbers (−0.04 after 12 months, with a 90 percent band of [−0.07, 0.00]), because a funds-rate innovation is short-lived. Adding a commodity price index, the standard remedy for the price puzzle (Sims, 1992; Ramey, 2016), leaves the two-year-yield response negative and significant after 12 months (−0.81; Appendix Table A14).

The first-stage F statistic of the proxy VAR is 4.5, in the range in which weak-instrument problems matter (Lewis and Mertens, 2026; Bauer et al., 2026), so we report exact Anderson–Rubin sets (Montiel Olea et al., 2021; Lunsford, 2015), which are valid whatever the strength of the instrument. These are the responses when the surprises are traced through the two-year yield alone.

3.2 The FCI model: baseline results

Figure 2 shows the baseline: the response of the unemployment gap and the core PCE price level to a policy move that raises the two-year yield 1 percentage point in the announcement window, from the model of Section 2.4 with the two-year yield and the FCI both measured relative to r^* (solid line, with 68 and 90 percent bands). The dashed line is the nominal two-year yield alone in the same model, and the dash-dotted line is the Cholesky VAR of Section 3.1. The units are the same: the end-of-month two-year yield moves 0.92 to 1.01 of the window move across the steps, so one percentage point in the window and one percentage point on the end-of-month yield in the monthly VARs are the same shock to within about 8 percent.

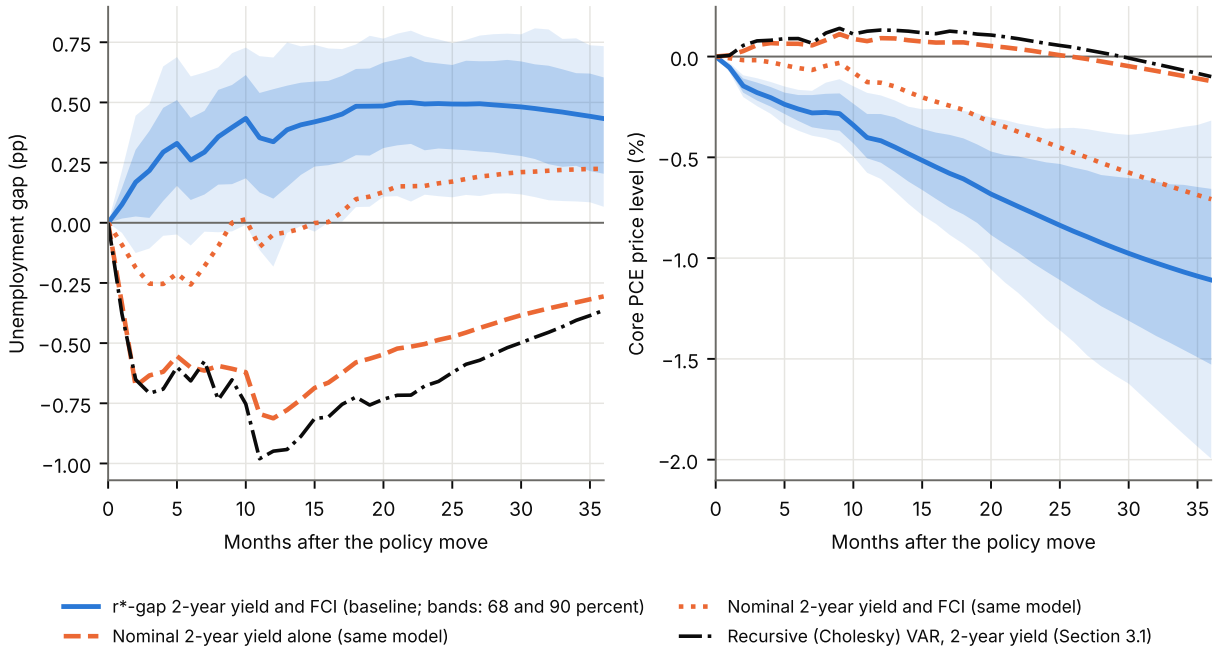


Figure 2: Responses to a policy move that raises the two-year yield 1 percentage point: the baseline (r^*-gap two-year yield and FCI; 12 lags; shaded bands 68 and 90 percent), the nominal two-year yield alone and with the FCI in the same model, and the Cholesky VAR of Section 3.1.

With the nominal yield alone the model gives what the standard methods give. Unemployment falls, to −0.81 percentage point after 12 months at its lowest, and is still −0.31 after 36 months; the price level hardly moves (within 0.15 percent of zero at every horizon in Table 3). The baseline is different. Unemployment rises, to 0.50 percentage point after 22 months, and the price level falls steadily, by 1.11 percent after 36 months. The dotted line shows how much of the difference the FCI alone makes: unemployment is 0.22 after 36 months and the price level falls by 0.71 percent.

The three steps are reported separately in Table 3 and Figure 3. In each step the two-year yield enters alone and then together with the FCI of the same step: nominal measures; real measures, net of expected inflation; and measures relative to r^*, net of expected inflation and of the Laubach–Williams estimate of r^* (Section 2.1 defines them). The tables show 12, 24 and 36 months; the monthly equations have 12 lags, fixed in advance, and Appendix Table A1 shows 3, 6 and 12 lags. The shocks and the monthly equations both start in January 1996, the first announcement for which all window measures are available (369 monthly observations).

Table 3: Baseline: 1996 to the present, all data in, no trend (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	12	369	−0.81*	−0.49*	−0.31	0.09	0.02	−0.12
				[−1.22, −0.39]	[−0.77, −0.17]	[−0.67, 0.04]	[−0.11, 0.32]	[−0.35, 0.44]	[−0.68, 0.52]
	nominal	12	369	−0.05	0.16	0.22	−0.13	−0.43	−0.71*
				[−0.54, 0.36]	[−0.14, 0.45]	[−0.07, 0.52]	[−0.38, 0.10]	[−0.86, 0.02]	[−1.30, −0.05]
Real	none	12	369	−0.60*	−0.33	−0.22	−0.03	−0.11	−0.21
				[−0.93, −0.22]	[−0.63, 0.01]	[−0.53, 0.14]	[−0.23, 0.16]	[−0.45, 0.21]	[−0.74, 0.26]
	real	12	369	0.09	0.23	0.21	−0.23*	−0.48*	−0.66*
				[−0.33, 0.49]	[−0.11, 0.57]	[−0.10, 0.54]	[−0.44, −0.02]	[−0.92, −0.04]	[−1.38, −0.03]
r*-gap	none	12	369	−0.29	−0.07	0.03	−0.21*	−0.44*	−0.69*
				[−0.91, 0.18]	[−0.68, 0.31]	[−0.58, 0.48]	[−0.40, −0.01]	[−0.82, −0.03]	[−1.35, −0.07]
	r*-gap	12	369	0.34	0.50*	0.43*	−0.42*	−0.81*	−1.11*
				[−0.18, 0.69]	[0.11, 0.76]	[0.07, 0.73]	[−0.61, −0.19]	[−1.30, −0.36]	[−2.00, −0.32]

Nominal, real and r*-gap 2-year yield; the FCI is Note A's index in the matching form in each case. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

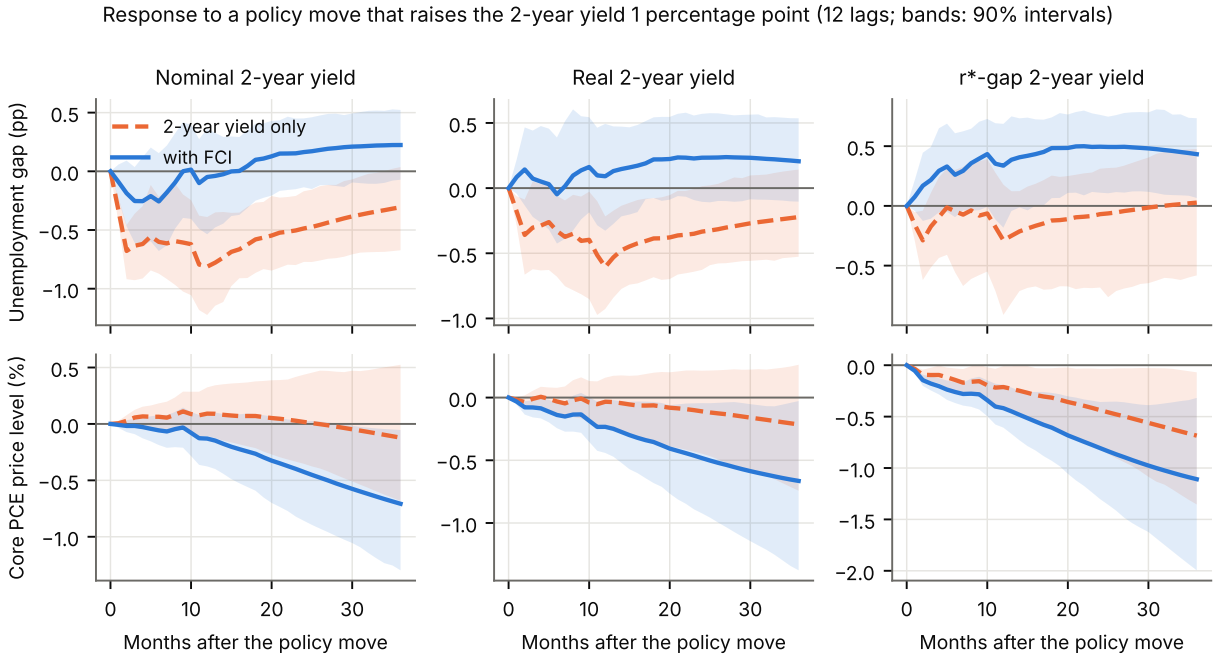


Figure 3: Responses to a policy move that raises the two-year yield 1 percentage point, without and with the FCI (12 lags). Bands: 90 percent intervals.

The two-year yield alone. At the nominal and real steps unemployment falls after a tightening, by 0.81 and 0.60 percentage point after 12 months (significant and significant), and the price level does not respond (0.02 and −0.11 percent after 24 months). Relative to r^*, the yield alone gives no rise in unemployment (−0.29 after 12 months) but does give a price fall of 0.44 percent after 24 months (significant).

With the FCI. Adding the FCI removes the fall in unemployment at every step: after 12 months the response is −0.05, 0.09 and 0.34 at the nominal, real and r^* steps, and after 24 months

0.16, 0.23 and 0.50. The price level falls at every step, by 0.43, 0.48 and 0.81 percent after 24 months and by 0.71, 0.66 and 1.11 percent after 36.

Reading across the steps. With the FCI, the unemployment response is positive after 24 and 36 months at every step but significant only at the r^* step, after 24 and 36 months. The price level falls at every step, significantly after 24 and 36 months at the real and r^* steps and after 36 months at the nominal step. The r^* step gives the largest price response and the only significant unemployment response; but the FCI alone already removes the fall in unemployment at every step.

3.3 The implied inflation-unemployment tradeoff

What matters for policy is the ratio of the price response to the unemployment response. Following Barnichon and Mesters (2021), we define the Phillips multiplier at horizon H as

$$\mathcal{P}_H = \frac{\tilde{P}_H}{\tilde{U}_H}, \qquad \tilde{U}_H = \frac{1}{12}\sum_{h=1}^{H} \tilde{U}_h, \tag{17}$$

where $\tilde{P}_H$ is the response of the core price level after H months, in percent, and $\tilde{U}_H$ is the cumulative response of the unemployment gap, in percentage-point years. Because the price level cumulates inflation, $\mathcal{P}_H$ is the cumulative change in inflation per unit of cumulative unemployment. Its inverse, $-1/\mathcal{P}_H$, is a sacrifice ratio: the percentage-point years of unemployment needed to lower the price level 1 percent. A more negative $\mathcal{P}_H$ means a steeper Phillips curve.

$\tilde{U}_H$ can be close to zero, and then the ratio is not well defined. So we form 90 percent sets for $\mathcal{P}_H$ in the spirit of Anderson and Rubin (1949): the set contains every value $\mathcal{P}$ for which the 90 percent bootstrap interval of $\tilde{P}_H - \mathcal{P}\,\tilde{U}_H$ contains zero. The sets are computed exactly, not on a grid, and can be unbounded, which shows when the data cannot pin down the tradeoff. Table 4 reports $\mathcal{P}_{24}$ and $\mathcal{P}_{36}$ for each step and for the main sample variations.

Table 4: Implied trade-off: Phillips multiplier κ_H (12 lags)

Case	FCI	$H = 24$ months	$H = 36$ months
Nominal	none	−0.01 [−0.32, 0.41]	0.08 [−0.25, 0.62]
	nominal	unbounded	unbounded
Real	none	0.14 [−0.24, 1.37]	0.21 [−0.22, 2.97]
	real	unbounded	unbounded
r*-gap	none	unbounded	unbounded
	r*-gap	−1.07 [−5.90, −0.41]	−0.90 [−3.27, −0.28]
Start 1973	r*-gap	−2.24 [−115.85, −1.24]	unbounded
2020 dropped	r*-gap	−0.95 [−1.81, −0.38]	−0.82 [−1.82, −0.26]
2020 dropped, start 1973	r*-gap	−1.49 [−2.48, −0.92]	−1.64 [−2.92, −0.95]
Ends 2019	r*-gap	−0.31 [−1.39, 0.31]	unbounded
Ends 2019, start 1973	r*-gap	−1.21 [−2.16, −0.58]	−1.58 [−3.60, −0.91]
r*-gap minus nominal, with FCI		[−1.56, 6.80]	[−1.27, 6.92]

κ_H = core PCE price level at month H (percent) per percentage-point year of cumulative unemployment-gap response (Barnichon and Mesters 2021); the sacrifice ratio is $-1/\kappa_H$. Point estimate and 90% confidence set (Anderson–Rubin type) in brackets, computed exactly (a wide grid refined by bisection), so a finite end is never an artifact of the grid. An infinite end means that the 90% interval of the cumulative unemployment response includes zero: the data then cannot rule out arbitrarily steep trade-offs on that side; "unbounded": both sides. Last row: paired difference over common announcement draws.

Only the baseline gives a tradeoff that can be measured. With the r^*-gap two-year yield and FCI, $\mathcal{P}_{24}$ is -1.07 with a 90 percent set of $[-5.90, -0.41]$ and $\mathcal{P}_{36}$ is -0.90 with $[-3.27, -0.28]$; both sets exclude zero. A sacrifice ratio is then 0.9 percentage-point years of unemployment per 1 percent of price level after 24 months. At the nominal and real steps with the FCI, and for the r^*-gap yield alone, the sets are unbounded: the cumulative unemployment response is too close to zero to pin down the ratio. With the nominal or real yield alone, the multiplier is close to zero and the sets include zero: a tightening that does not raise unemployment says nothing about the cost of lowering inflation. The multiplier is also negative, with a set that excludes zero, in four of the five sample variations in the lower rows of Table 4; the exception is the sample that starts in 1996 and ends in 2019. The difference between the r^*-gap and the nominal baseline is imprecise (last row).

[TO DO: Comparison with Barnichon and Mesters (2021) values: their numbers quoted in the earlier draft (-0.15 for 1990–2008; about -2 before 1990) are not verified in REFERENCE_CHECK; verify against the paper or drop.]

3.4 Financial conditions in the announcement window

The model assumes that policy reaches unemployment and prices through the two-year yield and the FCI, the broad index of the 10-year yield, equity valuations and the real dollar, weighted by their responses to policy news (Table 5). In the announcement window the FCI moves with the two-year yield: a policy move that raises the two-year yield 1 percentage point raises the FCI by 0.99 (in the same units). A model with the two-year yield alone omits this channel. Figure 4 shows the responses of the two-year yield and the FCI to the policy move in the baseline.

Table 5: Market-index weights from policy news

Component	Response to policy news	Weight	Share of a policy move
10-year Treasury yield (pp)	+0.53 (0.06)	+0.609	0.33
Equity valuation, 100 ln CAPE (%)	−7.78 (0.93)	−0.051	0.40
Real broad dollar, 100 ln (%)	+4.22 (0.53)	+0.065	0.28

Response of each 30-minute announcement-window change to the policy-news factor (scaled so the 2-year yield moves 1 pp), with robust standard errors; weights from the one-factor GLS of Note A; share = weight × response (the three shares sum to 1).

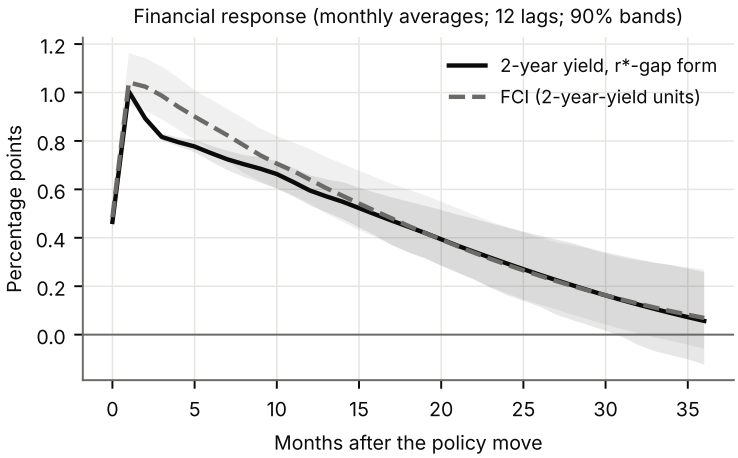


Figure 4: Response of the r^*-gap two-year yield and the FCI (in two-year-yield units) to a policy move that raises the two-year yield 1 percentage point in the window (monthly averages; 12 lags; 90 percent bands).

3.5 Proxy VARs with the FCI, and a simulation

The monthly VAR can be extended with the end-of-month FCI of the same step and estimated in two ways, with the same 1996 sample and announcements (equations, assumptions and references in baseline/lib/monthly.py). In the *proxy VAR* the immediate effect of the surprise on every variable is estimated from the monthly residuals and the instrument, as in the VARs without the FCI. In the *window-impact VAR* it is not estimated: the effect on the two-year yield and the FCI is taken from the announcement windows (the model's end-of-month response to a window surprise), and the effect on unemployment and inflation is set to zero within the month, the same restriction as in the main model. Table 6 gives the results at each step.

Table 6: Monthly VAR benchmark (12 lags)

					Unemployment gap (pp)			Price level (%)		
2-year	FCI	Identification	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	Proxy VAR	12	369	−0.91*	−0.61*	−0.32	0.03	−0.08	−0.26
					[−1.37, −0.42]	[−1.08, −0.05]	[−0.71, 0.17]	[−1.20, 0.97]	[−1.83, 1.25]	[−2.27, 1.25]
		Window impact	12	369	−0.90*	−0.63*	−0.35	0.13	0.06	−0.09
					[−1.45, −0.48]	[−1.17, −0.26]	[−0.95, 0.07]	[−0.10, 0.39]	[−0.41, 0.56]	[−0.78, 0.66]
	nominal	Proxy VAR	12	369	0.72	0.51	0.32	−0.67	−1.02	−1.17
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	12	369	−0.14	−0.09	−0.01	−0.13	−0.36	−0.57
					[−0.57, 0.37]	[−0.69, 0.45]	[−0.64, 0.47]	[−0.39, 0.08]	[−0.80, 0.07]	[−1.23, 0.06]
Real	none	Proxy VAR	12	369	−0.72*	−0.48*	−0.22	−0.12	−0.29	−0.46
					[−1.29, −0.35]	[−0.99, −0.07]	[−0.62, 0.15]	[−0.97, 0.71]	[−1.50, 0.85]	[−1.85, 0.80]
		Window impact	12	369	−0.65*	−0.46*	−0.23	−0.04	−0.16	−0.29
					[−1.18, −0.23]	[−1.01, −0.02]	[−0.77, 0.30]	[−0.27, 0.18]	[−0.62, 0.26]	[−1.01, 0.36]
	real	Proxy VAR	12	369	0.72	0.49	0.30	−0.72	−1.05	−1.12
					[−0.21, 24.27]	[−0.33, 19.69]	[−0.25, 12.47]	[−11.42, 0.16]	[−16.08, 0.22]	[−16.57, 0.33]
		Window impact	12	369	−0.02	−0.04	0.00	−0.26*	−0.49	−0.63
					[−0.51, 0.47]	[−0.61, 0.46]	[−0.60, 0.58]	[−0.63, −0.01]	[−1.11, 0.02]	[−1.53, 0.10]
r*-gap	none	Proxy VAR	12	369	−0.33	−0.10	0.06	−0.29	−0.57	−0.79
					[−0.98, 0.02]	[−0.64, 0.24]	[−0.30, 0.32]	[−0.97, 0.48]	[−1.51, 0.47]	[−1.83, 0.31]
		Window impact	12	369	−0.19	−0.03	0.09	−0.27*	−0.51*	−0.69*
					[−0.70, 0.31]	[−0.53, 0.43]	[−0.27, 0.61]	[−0.51, −0.05]	[−0.99, −0.08]	[−1.38, −0.07]
	r*-gap	Proxy VAR	12	369	0.95*	0.73	0.47	−0.84*	−1.24*	−1.35*
					[0.01, 5.87]	[−0.04, 4.55]	[−0.05, 2.89]	[−3.17, −0.15]	[−4.39, −0.24]	[−4.51, −0.24]
		Window impact	12	369	0.33	0.31	0.25	−0.45*	−0.77*	−0.94*
					[−0.19, 0.77]	[−0.18, 0.90]	[−0.22, 0.81]	[−0.70, −0.22]	[−1.27, −0.33]	[−1.74, −0.36]

Monthly VAR in the unemployment gap, core PCE inflation, the end-of-month 2-year yield and, when included, the end-of-month FCI (same step), 1996 to September 2026, no trend, nothing dropped. Proxy VAR: Nakamura–Steinsson surprise as instrument; sets are exact 90% Anderson–Rubin sets, "unbounded" when the instrument cannot reject large effects. Window impact: impact vector taken from the daily model (the end-of-month response to a window surprise); 90% bootstrap intervals. Unit in both: +1 pp on the 2-year yield (the end-of-month response to a +1 pp window surprise is 0.92 to 1.01, so the two units are the same up to 8%). * = 90% interval excludes zero.

The first-stage F statistic of the proxy VAR with the FCI is 4.4 at the r^* step, and Appendix Table A15 reports it for each step. The proxy also implies an end-of-month response of the FCI of 1.44 per percentage point of the two-year yield, against 1.03 in the window; the 12-month response of unemployment at the r^* step is 0.95 with a 90 percent set of [0.01, 5.87], and at the nominal step with the FCI the set is $[-\infty, +\infty]$. The window-impact VAR is close to the main model: with the r^*-gap yield and FCI, unemployment rises 0.33 percentage point after 12 months and the price level falls 0.77 percent after 24, and without the FCI it does not rise (−0.19 after 12 months).

If the model were the true process, how noisy would the proxy VAR be in samples like ours? We simulate 1,000 samples of 369 months from the estimated r^*-gap model, with the data's calendar and announcement days, resample the daily and monthly innovations and the surprises, and apply the monthly estimators to the end-of-month series of each sample (Table 7).

Table 7: A monthly proxy VAR when the r*-gap model is the true process

Estimate	Data	True	Simulated median	5th–95th percentile
A. Proxy VAR with the FCI				
First-stage F	4.4		11.1	[2.5, 27.2]
Unemployment gap, 12 m	0.95	0.33	0.35	[−0.31, 1.26]
Price level, 24 m	−1.24	−0.80	−0.69	[−1.91, 0.17]
B. Proxy VAR without the FCI				
Unemployment gap, 12 m	−0.33	0.33	0.02	[−0.70, 0.70]
C. Monthly VAR with the window impact and the FCI				
Unemployment gap, 12 m	0.33	0.34	0.35	[−0.10, 0.80]

1,000 samples of 369 months generated by the estimated r*-gap model with the FCI (12 lags), with the data's calendar and announcement days; policy reaches unemployment and prices only through the 2-year yield and the FCI. Data: the 1996 sample, 12 lags, r*-gap measures; end-of-month series in the monthly VAR. True: the model's response per +1 pp on the 2-year yield, in the window (panel C) and on the end-of-month yield (panels A and B; the window response divided by the model's end-of-month response of 0.92–1.01). Simulated: median and 5th–95th percentiles of the estimates.

Under the model the true response of unemployment after 12 months is 0.33 percentage point. The proxy VAR with the FCI is centred on it (median 0.35) but very noisy: the 5th to 95th percentile range is [−0.31, 1.26], and the first-stage F is below the data value (4.4) in 14 percent of samples, with a median of 11.1. Our data value 0.95 lies inside that range. Without the FCI the proxy VAR is biased toward zero (median 0.02 against the truth 0.33); the data value −0.33 is within the simulated range. The window-impact VAR recovers the truth (median 0.35) with a narrower range. The comparison is conditional on the model being the true process: if it is, the monthly proxy VAR is a noisy way to estimate the effect, and a first-stage statistic and responses like those in the data are not unusual.

4 Robustness

Each variation changes one feature of the baseline and keeps the rest: the shocks are the same 252 announcements, the monthly equations have 12 lags, and responses are to a policy move that raises the two-year yield 1 percentage point in the announcement window. Every table shows the two-year yield without and with the FCI of the same step, and Appendix tables give 3, 6 and 12 lags.

4.1 The sample

Three variations change the sample. In the first, the monthly equations start in 1973 rather than 1996; the months before 1990, for which there are no daily data, use monthly series in place of the daily block, the daily block is still estimated from 1996, and the shocks are still the announcements from 1996. In the second, March to December 2020 are dropped from the monthly and daily equations (2020 values still enter as lags). In the third, all equations end in December 2019. Each is run with the monthly equations starting in 1996 and in 1973. Table 8 shows the r^*-gap two-year yield without and with the r^*-gap FCI.

Table 8: Robustness 1: sample (r*-gap 2-year yield and FCI) (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Start 1973	none	12	633	−0.05	−0.01	0.00	−0.22*	−0.50*	−0.79*
				[−0.22, 0.13]	[−0.20, 0.17]	[−0.19, 0.24]	[−0.36, −0.11]	[−0.79, −0.24]	[−1.26, −0.31]
	r*-gap	12	633	0.18	0.14	0.13	−0.27*	−0.63*	−0.98*
				[−0.03, 0.38]	[−0.03, 0.31]	[−0.08, 0.34]	[−0.45, −0.12]	[−0.97, −0.30]	[−1.54, −0.44]
2020 dropped	none	12	359	−0.10	−0.21	−0.25	−0.20	−0.32	−0.44
				[−0.33, 0.10]	[−0.66, 0.15]	[−0.84, 0.21]	[−0.38, 0.00]	[−0.76, 0.07]	[−1.15, 0.22]
	r*-gap	12	359	0.41*	0.45*	0.40*	−0.38*	−0.69*	−0.94*
				[0.23, 0.65]	[0.16, 0.75]	[0.03, 0.78]	[−0.58, −0.15]	[−1.15, −0.25]	[−1.71, −0.30]
2020 dropped, start 1973	none	12	623	0.17*	0.15*	0.05	−0.28*	−0.64*	−0.97*
				[0.06, 0.29]	[0.00, 0.31]	[−0.14, 0.22]	[−0.40, −0.16]	[−0.89, −0.37]	[−1.35, −0.59]
	r*-gap	12	623	0.33*	0.24*	0.15	−0.32*	−0.76*	−1.14*
				[0.20, 0.44]	[0.08, 0.40]	[−0.02, 0.32]	[−0.47, −0.16]	[−1.07, −0.42]	[−1.73, −0.58]
Ends 2019	none	12	288	−0.15	−0.24	−0.26	0.08	0.06	0.03
				[−0.34, 0.15]	[−0.61, 0.18]	[−0.77, 0.25]	[−0.05, 0.20]	[−0.11, 0.25]	[−0.20, 0.27]
	r*-gap	12	288	0.30*	0.13	0.00	−0.08	−0.14	−0.21
				[0.10, 0.48]	[−0.15, 0.39]	[−0.36, 0.36]	[−0.23, 0.08]	[−0.42, 0.10]	[−0.60, 0.11]
Ends 2019, start 1973	none	12	552	0.19*	0.18*	0.08	−0.21*	−0.49*	−0.82*
				[0.06, 0.34]	[0.02, 0.40]	[−0.13, 0.36]	[−0.34, −0.06]	[−0.82, −0.15]	[−1.39, −0.23]
	r*-gap	12	552	0.27*	0.16*	0.09	−0.22*	−0.50*	−0.84*
				[0.15, 0.40]	[0.02, 0.36]	[−0.10, 0.33]	[−0.39, −0.07]	[−0.90, −0.21]	[−1.50, −0.38]

Start 1973: monthly equations from 1973, with monthly series in place of daily data before 1990 (lib/model.py). 2020 dropped: March to December 2020 removed from the monthly and daily equations (2020 values still enter as lags). Ends 2019: all equations, monthly and daily. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

With the FCI, the unemployment response is positive after 12 months in every variation and significant in all but the 1973 start (0.18 after 12 months, not significant). Dropping 2020 leaves it significant: 0.41 after 12 months from 1996 and 0.33 from 1973, against 0.34 in the baseline. The price response is significant at every horizon in all the variations that start in 1973, with the FCI and without it (for example −0.63 and −0.50 after 24 months from a 1973 start). From 1996 it is significant when 2020 is dropped (−0.69 after 24 months) but is smaller and not significant when the sample ends in 2019 (−0.14), where the unemployment response is significant after 12 months only.

4.2 The r^* adjustment and trends

Two variations change how slow-moving trends are handled. The first replaces the Laubach–Williams estimate of r^* with Kiley's, in the two-year yield and in the FCI alike. The second uses no r^* measure: it keeps the real two-year yield and the real FCI, and adds a natural cubic spline in time to every monthly equation, with knots at the sample ends and at the NBER business-cycle peaks inside the sample. Table 9 gives the results.

Table 9: Robustness 2: the r* adjustment and trends (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	3
Kiley r*	none	12	369	−0.53*	−0.32	−0.23	−0.09	−0.10	−
				[−0.93, −0.14]	[−0.64, 0.00]	[−0.59, 0.17]	[−0.27, 0.08]	[−0.46, 0.20]	[−0.73,
	r*-gap (Kiley)	12	369	0.39	0.28	−0.01	−0.25*	−0.18	−
				[−0.02, 0.77]	[−0.07, 0.80]	[−0.38, 0.46]	[−0.48, −0.03]	[−0.72, 0.36]	[−0.93,
Kiley r*, start 1973	none	12	633	−0.11	−0.05	−0.03	−0.18*	−0.35*	−0
				[−0.25, 0.03]	[−0.21, 0.12]	[−0.19, 0.14]	[−0.30, −0.06]	[−0.62, −0.11]	[−0.91, −
	r*-gap (Kiley)	12	633	0.00	−0.04	−0.03	−0.20*	−0.37*	−0
				[−0.14, 0.19]	[−0.17, 0.07]	[−0.16, 0.09]	[−0.33, −0.06]	[−0.63, −0.12]	[−0.88, −
Real, spline trend	none	12	369	−0.64*	−0.34	−0.20	0.03	−0.06	−
				[−1.07, −0.26]	[−0.75, 0.06]	[−0.61, 0.28]	[−0.18, 0.20]	[−0.39, 0.29]	[−0.77,
	real	12	369	0.19	0.28	0.17	−0.14	−0.25	−
				[−0.27, 0.63]	[−0.18, 0.78]	[−0.21, 0.62]	[−0.42, 0.08]	[−0.87, 0.21]	[−1.23,
Real, spline trend, start 1973	none	12	633	−0.10	0.08	0.11	−0.09	−0.25*	−0
				[−0.29, 0.06]	[−0.22, 0.29]	[−0.18, 0.34]	[−0.18, 0.01]	[−0.42, −0.02]	[−0.64, −
	real	12	633	0.18	0.18	0.11	−0.17*	−0.39*	−0
				[0.00, 0.37]	[−0.02, 0.39]	[−0.04, 0.27]	[−0.29, −0.03]	[−0.62, −0.15]	[−0.85, −

Kiley r*: 2-year minus expected inflation minus Kiley's r*, with Note A's index built with the same r*. Real, spline trend: real 2-year and real FCI with a natural cubic spline trend (knots at the sample ends and at NBER business-cycle peaks inside the sample) in every monthly equation, in place of any r* measure. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

With Kiley's r^* and the FCI, unemployment rises 0.39 after 12 months (not significant) and 0.28 after 24 (not significant), against a fall of 0.53 without the FCI (significant). The price response is smaller than in the baseline: −0.25 after 12 months (significant) and −0.18 after 24 (not significant). From 1973 the price response is significant at every horizon (−0.37 after 24 months), with the unemployment response small. With real measures and the spline, the FCI again removes the fall in unemployment: −0.64 after 12 months without it (significant) and 0.19 with it (not significant). From 1996 the responses with the FCI have the standard signs but are imprecise. From 1973 the price response is significant at every horizon (−0.39 after 24 months), and the unemployment response is positive (0.18 after 12 months) but not significant.

4.3 Credit conditions

Credit is the most important channel the baseline leaves out (Gilchrist and Zakrajšek, 2012; Caldara and Herbst, 2019). The variation adds the monthly excess bond premium as a third monthly variable, treated like unemployment and inflation, so that the premium responds to the financial variables and they enter its equation as well (Table 10).

Table 10: Robustness 3: excess bond premium (r*-gap) (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
EBP added	none	12	369	−0.08	0.08	0.16	−0.27*	−0.53*	−0.78*
				[−0.42, 0.27]	[−0.38, 0.44]	[−0.43, 0.62]	[−0.41, −0.06]	[−0.88, −0.09]	[−1.38, −0.15]
	r*-gap	12	369	0.27	0.52*	0.42	−0.39*	−0.78*	−1.10*
				[−0.24, 0.79]	[0.01, 0.93]	[−0.08, 0.86]	[−0.59, −0.16]	[−1.21, −0.32]	[−1.81, −0.30]
EBP added, start 1973	none	12	633	−0.03	−0.01	0.01	−0.22*	−0.50*	−0.78*
				[−0.17, 0.14]	[−0.18, 0.16]	[−0.19, 0.18]	[−0.38, −0.11]	[−0.83, −0.23]	[−1.34, −0.29]
	r*-gap	12	633	0.06	0.03	0.04	−0.24*	−0.57*	−0.89*
				[−0.12, 0.24]	[−0.19, 0.23]	[−0.17, 0.25]	[−0.40, −0.08]	[−0.92, −0.22]	[−1.43, −0.39]

The excess bond premium is a third monthly variable, treated like unemployment and inflation (u, inflation, EBP system); responses shown for u and the price level. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

With the r^*-gap FCI, unemployment rises 0.27 after 12 months and 0.52 after 24 (significant at 24 months), and the price level falls 0.78 percent after 24 months (significant). Without the FCI the unemployment response is close to zero (−0.08 after 12 months), nearer to zero than in the baseline without the FCI (−0.29). One reading is that the premium, like the FCI, absorbs policy-driven moves in risk premia (Bauer et al., 2026); credit conditions then give a partial alternative to the FCI for the unemployment response, though not for the price response, which is significant without the FCI as well. From 1973 the unemployment response is close to zero with and without the FCI, and the price response is significant in both.

4.4 The weights of the FCI

The baseline assumes that the three parts of the FCI, the 10-year yield, equity valuation and the dollar, affect unemployment and prices in proportion to the weights with which they respond to policy news (Section 2.3). The variation relaxes this. The three weighted components enter the daily and monthly equations separately, in the step of the two-year yield, in place of the FCI, and each has its own window impact; the three impacts add up to the impact of the FCI. To keep the number of parameters reasonable, the daily block is restricted: each component's equation contains its own 1-, 5-, 22- and 66-day averages and those of the two-year yield, plus the monthly macro variables, and the two-year yield's equation its own averages and the macro variables. That is 48 coefficients in all, against 26 in the baseline daily block with the FCI. The monthly equations have 12 lags and start in 1996 or, as a further variation, in 1973. Table 11 shows the responses.

Table 11: Robustness 4: the FCI replaced by its three weighted components (12 lags)

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	components	12	369	0.03	0.25	0.09	0.03	0.21	0.42
				[−0.49, 0.56]	[−0.21, 0.70]	[−0.31, 0.41]	[−0.25, 0.29]	[−0.36, 0.68]	[−0.54, 1.17]
Real	components	12	369	0.17	0.29	−0.03	−0.06	0.14	0.41
				[−0.32, 0.63]	[−0.11, 0.65]	[−0.33, 0.23]	[−0.29, 0.18]	[−0.36, 0.61]	[−0.32, 1.08]
r*-gap	components	12	369	0.20	0.22	−0.09	−0.14	0.06	0.35
				[−0.20, 0.69]	[−0.14, 0.61]	[−0.42, 0.21]	[−0.36, 0.04]	[−0.40, 0.47]	[−0.40, 0.98]
Nominal, start 1973	components	12	633	−0.28	−0.24	−0.26	0.10	0.45	0.92
				[−0.58, 0.11]	[−0.58, 0.03]	[−0.63, 0.02]	[−0.26, 0.38]	[−0.33, 1.13]	[−0.33, 2.03]
Real, start 1973	components	12	633	−0.11	−0.14	−0.15	−0.14	−0.05	0.16
				[−0.52, 0.23]	[−0.45, 0.15]	[−0.50, 0.13]	[−0.47, 0.14]	[−0.73, 0.62]	[−0.96, 1.30]
r*-gap, start 1973	components	12	633	0.17	0.16	0.09	−0.35*	−0.60*	−0.80
				[−0.14, 0.55]	[−0.12, 0.47]	[−0.14, 0.44]	[−0.62, −0.14]	[−1.31, −0.04]	[−1.90, 0.13]

Components: the 10-year yield, equity valuation and real dollar, each multiplied by its weight, enter separately in the daily and monthly equations in the same step as the 2-year yield, each with its own window impact (the three impacts sum to the FCI's). Baseline rows without the FCI are in the baseline table. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Freeing the weights adds many coefficients and the intervals widen. At the r^* step the responses keep the pattern of the baseline. Unemployment rises, by 0.20 after 12 months and 0.22 after 24 from 1996, and by 0.17 and 0.16 from 1973, in each case within the interval of the baseline response. From 1973 the price level falls significantly, 0.35 percent after 12 months and 0.60 percent after 24, close to the response with the FCI (−0.63 after 24 months). From 1996 the price response is not distinguishable from zero (0.06 after 24 months, with an interval of [−0.40, 0.47]). Table 12 tests the restriction that the coefficients on the three components are equal at each lag, with bootstrap p-values. It is not rejected in the unemployment equation at any step. In the inflation equation it is rejected from 1996 (0.002 at the r^* step) and not from 1973 (0.426).

Table 12: Test of the proportional-weights restriction (12 lags)

	Unemployment gap equation		Core PCE inflation equation	
Step	Wald	Bootstrap p	Wald	Bootstrap p
Nominal	8.9	0.970	41.9	0.044
Real	7.0	0.991	42.9	0.037
r*-gap	21.7	0.336	53.9	0.002
Nominal, start 1973	24.5	0.922	26.0	0.576
Real, start 1973	22.2	0.869	32.4	0.283
r*-gap, start 1973	33.4	0.344	27.9	0.426

Wald statistic (heteroskedasticity-robust, HC1) for the restriction that the coefficients on the three weighted components (10-year yield, equity valuation, dollar) are equal at each lag (24 restrictions), in each monthly equation, and the bootstrap p-value (999 residual draws under the restriction); monthly equations from 1996 (first three rows) and from 1973 (last three), no trend, 2020 in. The unrestricted equation is the one used in Robustness 4.

5 Conclusion and policy

5.1 Implications for policy

Weak responses may reflect how the stance was measured. With the nominal two-year yield alone, the surprises give the responses that standard methods give after the early 1980s: unemployment falls and prices do not respond. With broad financial conditions and the two-year yield measured relative to r^*, the responses have the standard signs. A policy move that raises the two-year yield 1 percentage point raises the unemployment gap 0.50 percentage point after two years and lowers the core price level 0.81 percent. This does not show that the weak responses are an artifact; it shows that they are not a feature of the surprises themselves.

Judge the stance by financial conditions relative to r^*. In these estimates policy reaches the economy through broad financial conditions, and the right yardstick for them is the neutral rate r^*. This supports the practice of monitoring financial conditions when setting policy, as the FOMC did in 2023 (Powell, 2023), and it explains why the same policy rate can be more or less restrictive at different times. One caution applies: r^* is uncertain, and our intervals hold it fixed. The Laubach–Williams estimates we use are one-sided, but their parameters are estimated on the full sample, so they are not what a policymaker could have known in real time.

The cost of disinflation. Measured relative to r^*, lowering the core price level 1 percent through tighter financial conditions costs about 0.9 percentage-point years of unemployment over two years (for example, unemployment about half a point higher for two years). With the nominal or real FCI, or with the yield alone, the data cannot pin the cost down. The estimate itself is imprecise: the 90 percent set for the multiplier runs from -5.90 to -0.41.

What this paper does not do. Our estimates are effects of surprises. They do not by themselves say how much of the 2022–24 disinflation, or of the rise in unemployment, was caused by the tightening; that requires a decomposition of the whole episode, which we leave to future work. We also do not test whether tightenings and easings, or periods of high and low inflation, have different effects.

5.2 Conclusion

Interest-rate surprises around Federal Reserve announcements measure policy shocks well, at least the part of policy news that moves interest rates. The hard part has been tracing their effects on the economy. We keep the surprise as it is and change how its effects are traced: through broad financial conditions rather than one interest rate, and relative to the neutral rate r^*. Traced this way, the responses of unemployment and prices have the standard signs, where the nominal two-year yield alone and the standard methods give the wrong ones. The implied tradeoff between inflation and unemployment is of conventional size.

The two parts play different roles. Within our design, the FCI, with its response measured in the window, removes the wrong-signed unemployment response at every step. Measuring conditions relative to r^* gives the significant unemployment and price responses and the tradeoff.

These findings complement work on information effects, predictable surprises, timing and policy-driven risk appetite. That last line of work and ours agree that stock prices and the dollar carry much of the effect of policy; they differ in whether those moves are a separate shock or part of financial conditions. A natural next step is to trace cleaned or separated surprises through our approach.

Our results have limits. They rest on the assumption that the economy responds to financial conditions the same way whatever moved them, and our intervals do not include uncertainty about r^* or the FCI weights.

The broader lesson is that the fall in r^* over the past three decades is more than a fact about the level of interest rates. Unless rates are measured relative to r^*, that fall also makes the effects of policy, and the cost of lowering inflation, harder to measure.

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[TO DO: Appendix: the 3/6/12-lag tables now come from baseline/. Still to check: the data-sources table and the construction text against baseline/data/SOURCES.md (unverified items: EBP file origin, r* vintage, daily CAPE and dollar file, DFF and PPIACO copies).]

A Additional tables

Table A14: Standard methods with a commodity price index in the first block

			Unemployment gap (pp)			Price level (%)	
Method	Sample	Obs	12 m	24 m	36 m	12 m	24 m
Cholesky + commodity prices, 2-year yield	1996–2026	369	−0.81*	−0.44	−0.13	0.11	0.06
			[−1.32, −0.39]	[−0.95, 0.06]	[−0.67, 0.46]	[−0.13, 0.39]	[−0.45, 0.63] [-
Cholesky + commodity prices, federal funds rate	1984–2007	288	−0.03	−0.02	−0.01	−0.02	−0.04
			[−0.08, 0.00]	[−0.07, 0.02]	[−0.05, 0.03]	[−0.07, 0.02]	[−0.12, 0.04] [-

Monthly data, 12 lags, constant, no trend; unemployment gap, core PCE inflation and the end-of-month policy rate (the 2-year Treasury yield, or the effective federal funds rate where stated), policy rate ordered last in the Cholesky VARs; price level = cumulated inflation/12. The log PPI for all commodities (monthly average) is added to the system and ordered before the policy rate. Unit: +1 pp on the policy rate on impact. Brackets: 90% intervals (residual bootstrap for the Cholesky VARs; exact Anderson–Rubin sets for the proxy VAR). * = excludes zero. Specification: baseline/lib/litstyle.py.

Table A15: Monthly proxy VAR: first-stage strength and impact on the FCI

2-year	FCI	F (3 lags)	F (6)	F (12)	Proxy impact on FCI	Model impact on FCI
Nominal	none	5.0	4.6	4.5		
	nominal	2.7	2.1	1.6	1.73	1.13
Real	none	9.0	7.3	6.9		
	real	6.0	4.2	3.0	1.50	1.03
r*-gap	none	9.1	8.2	8.0		
	r*-gap	8.4	6.7	4.4	1.44	1.03

F: robust first-stage statistic of the policy-variable residual on the instrument (instrument months only, 1996–2026). Impact on FCI: end-of-month response of the FCI per +1 pp on the end-of-month 2-year yield, estimated from monthly residuals (proxy VAR, 12 lags) and implied by the daily model.

B Construction and estimation details

Events. We use the 252 FOMC announcements from January 1996 to July 2026 with all the window measures available in the U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). As in the companion note, we leave out unscheduled announcements with no change in the policy rate, and March–April 2020. The one later announcement in the database, September 2026, is left out because too few later days are available. The DXY index used in the window covers fewer currencies than the broad dollar used in the daily and monthly index, which we treat as a measurement difference.

From monthly to daily. Expected inflation and r^* are not available daily, and they move slowly. We take the monthly gap between R_t^v in equation (8) and y_t^2, interpolate it linearly between

Table A1: Baseline: 1996 to the present, all data in, no trend

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	3	369	−0.59*	−0.45*	−0.36	0.11	0.08	0.05
				[−0.80, −0.28]	[−0.75, −0.09]	[−0.68, 0.01]	[−0.06, 0.28]	[−0.19, 0.38]	[−0.34, 0.49]
		6	369	−0.55*	−0.42*	−0.30	0.07	0.01	−0.10
				[−0.79, −0.28]	[−0.69, −0.10]	[−0.64, 0.06]	[−0.14, 0.27]	[−0.42, 0.35]	[−0.72, 0.41]
		12	369	−0.81*	−0.49*	−0.31	0.09	0.02	−0.12
				[−1.22, −0.39]	[−0.77, −0.17]	[−0.67, 0.04]	[−0.11, 0.32]	[−0.35, 0.44]	[−0.68, 0.52]
	nominal	3	369	−0.02	0.07	0.05	−0.10	−0.27	−0.37
				[−0.28, 0.28]	[−0.16, 0.35]	[−0.12, 0.25]	[−0.30, 0.08]	[−0.64, 0.05]	[−0.88, 0.08]
		6	369	0.01	0.17	0.17	−0.14	−0.39	−0.60
				[−0.26, 0.26]	[−0.12, 0.44]	[−0.09, 0.42]	[−0.41, 0.12]	[−0.92, 0.08]	[−1.33, 0.00]
		12	369	−0.05	0.16	0.22	−0.13	−0.43	−0.71*
				[−0.54, 0.36]	[−0.14, 0.45]	[−0.07, 0.52]	[−0.38, 0.10]	[−0.86, 0.02]	[−1.30, −0.05]
Real	none	3	369	−0.34*	−0.33	−0.29	−0.05	−0.07	−0.09
				[−0.54, −0.04]	[−0.54, 0.04]	[−0.55, 0.06]	[−0.24, 0.10]	[−0.39, 0.17]	[−0.53, 0.27]
		6	369	−0.33	−0.29	−0.22	−0.01	−0.09	−0.19
				[−0.61, 0.00]	[−0.58, 0.07]	[−0.56, 0.12]	[−0.19, 0.17]	[−0.40, 0.20]	[−0.59, 0.22]
		12	369	−0.60*	−0.33	−0.22	−0.03	−0.11	−0.21
				[−0.93, −0.22]	[−0.63, 0.01]	[−0.53, 0.14]	[−0.23, 0.16]	[−0.45, 0.21]	[−0.74, 0.26]
	real	3	369	0.12	0.06	0.00	−0.20	−0.31	−0.37
				[−0.19, 0.37]	[−0.22, 0.30]	[−0.21, 0.13]	[−0.38, 0.02]	[−0.60, 0.03]	[−0.72, 0.05]
		6	369	0.14	0.19	0.14	−0.20*	−0.42*	−0.60*
				[−0.11, 0.41]	[−0.10, 0.47]	[−0.09, 0.36]	[−0.40, −0.01]	[−0.77, −0.07]	[−1.08, −0.05]
		12	369	0.09	0.23	0.21	−0.23*	−0.48*	−0.66*
				[−0.33, 0.49]	[−0.11, 0.57]	[−0.10, 0.54]	[−0.44, −0.02]	[−0.92, −0.04]	[−1.38, −0.03]
r*-gap	none	3	369	−0.17	−0.16	−0.14	−0.12	−0.15	−0.17
				[−0.43, 0.19]	[−0.50, 0.24]	[−0.45, 0.21]	[−0.26, 0.05]	[−0.43, 0.15]	[−0.55, 0.26]
		6	369	−0.09	−0.06	−0.01	−0.16	−0.35	−0.53
				[−0.42, 0.20]	[−0.49, 0.32]	[−0.49, 0.40]	[−0.36, 0.04]	[−0.74, 0.07]	[−1.09, 0.02]
		12	369	−0.29	−0.07	0.03	−0.21*	−0.44*	−0.69*
				[−0.91, 0.18]	[−0.68, 0.31]	[−0.58, 0.48]	[−0.40, −0.01]	[−0.82, −0.03]	[−1.35, −0.07]
	r*-gap	3	369	0.26	0.13	0.01	−0.20	−0.25	−0.27
				[−0.14, 0.55]	[−0.22, 0.39]	[−0.26, 0.20]	[−0.43, 0.00]	[−0.65, 0.09]	[−0.78, 0.18]
		6	369	0.34*	0.36*	0.26	−0.33*	−0.63*	−0.87*
				[0.06, 0.55]	[0.03, 0.64]	[−0.05, 0.51]	[−0.55, −0.11]	[−1.14, −0.21]	[−1.77, −0.26]
		12	369	0.34	0.50*	0.43*	−0.42*	−0.81*	−1.11*
				[−0.18, 0.69]	[0.11, 0.76]	[0.07, 0.73]	[−0.61, −0.19]	[−1.30, −0.36]	[−2.00, −0.32]

Nominal, real and r*-gap 2-year yield; the FCI is the matching index of Note A in each case. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A2: Robustness 1a: monthly equations start in 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	642	0.01	0.05	0.05	−0.09	−0.15	−0.20
				[−0.10, 0.20]	[−0.11, 0.26]	[−0.10, 0.28]	[−0.23, 0.06]	[−0.48, 0.22]	[−0.72, 0.39]
		6	639	0.00	0.03	0.03	−0.12	−0.31	−0.53
				[−0.13, 0.12]	[−0.14, 0.18]	[−0.15, 0.18]	[−0.24, 0.02]	[−0.60, 0.02]	[−1.02, 0.01]
		12	633	−0.05	−0.01	0.00	−0.22*	−0.50*	−0.79*
				[−0.22, 0.13]	[−0.20, 0.17]	[−0.19, 0.24]	[−0.36, −0.11]	[−0.79, −0.24]	[−1.26, −0.31]
	r*-gap	3	642	0.06	0.00	−0.04	−0.08	−0.16	−0.27
				[−0.05, 0.17]	[−0.16, 0.16]	[−0.21, 0.15]	[−0.23, 0.07]	[−0.53, 0.17]	[−0.81, 0.23]
		6	639	0.09	0.05	0.05	−0.14	−0.35*	−0.61*
				[−0.06, 0.29]	[−0.15, 0.27]	[−0.18, 0.26]	[−0.30, 0.02]	[−0.68, −0.03]	[−1.17, −0.14]
		12	633	0.18	0.14	0.13	−0.27*	−0.63*	−0.98*
				[−0.03, 0.38]	[−0.03, 0.31]	[−0.08, 0.34]	[−0.45, −0.12]	[−0.97, −0.30]	[−1.54, −0.44]

r*-gap only. Pre-1990 months use monthly series in place of the daily data (see lib/model.py); shocks are the 252 announcements of 1996 on. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A3: Robustness 1b: 2020 dropped from all equations, start 1996

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	359	0.13	0.22	0.26	−0.18*	−0.28	−0.35
				[−0.01, 0.29]	[−0.01, 0.51]	[−0.02, 0.63]	[−0.37, −0.02]	[−0.65, 0.04]	[−0.83, 0.13]
		6	359	0.04	0.05	0.07	−0.15	−0.30	−0.44
				[−0.21, 0.25]	[−0.35, 0.42]	[−0.38, 0.54]	[−0.34, 0.05]	[−0.67, 0.07]	[−0.98, 0.15]
		12	359	−0.10	−0.21	−0.25	−0.20	−0.32	−0.44
				[−0.33, 0.10]	[−0.66, 0.15]	[−0.84, 0.21]	[−0.38, 0.00]	[−0.76, 0.07]	[−1.15, 0.22]
	r*-gap	3	359	0.37*	0.44*	0.39*	−0.22*	−0.35	−0.47
				[0.20, 0.55]	[0.19, 0.75]	[0.08, 0.76]	[−0.46, −0.03]	[−0.80, 0.04]	[−1.09, 0.07]
		6	359	0.38*	0.44*	0.43	−0.30*	−0.57*	−0.80
				[0.11, 0.57]	[0.05, 0.80]	[−0.04, 0.94]	[−0.56, −0.06]	[−1.15, −0.04]	[−1.76, 0.02]
		12	359	0.41*	0.45*	0.40*	−0.38*	−0.69*	−0.94*
				[0.23, 0.65]	[0.16, 0.75]	[0.03, 0.78]	[−0.58, −0.15]	[−1.15, −0.25]	[−1.71, −0.30]

r*-gap only. March to December 2020 dropped from the monthly and the daily equations. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A4: Robustness 1c: 2020 dropped from all equations, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	632	0.06	0.11	0.13	−0.11	−0.20	−0.27
				[−0.02, 0.14]	[−0.04, 0.24]	[−0.04, 0.30]	[−0.26, 0.03]	[−0.53, 0.13]	[−0.80, 0.22]
		6	629	0.11*	0.10	0.05	−0.16*	−0.39*	−0.62*
				[0.01, 0.22]	[−0.06, 0.25]	[−0.15, 0.22]	[−0.30, −0.03]	[−0.70, −0.05]	[−1.13, −0.12]
		12	623	0.17*	0.15*	0.05	−0.28*	−0.64*	−0.97*
				[0.06, 0.29]	[0.00, 0.31]	[−0.14, 0.22]	[−0.40, −0.16]	[−0.89, −0.37]	[−1.35, −0.59]
	r*-gap	3	632	0.12*	0.09	0.06	−0.09	−0.25*	−0.48*
				[0.03, 0.22]	[−0.06, 0.25]	[−0.11, 0.23]	[−0.25, 0.01]	[−0.59, −0.02]	[−0.94, −0.09]
		6	629	0.27*	0.15	0.05	−0.20*	−0.49*	−0.80*
				[0.16, 0.39]	[−0.02, 0.31]	[−0.15, 0.26]	[−0.40, −0.07]	[−0.93, −0.18]	[−1.41, −0.32]
		12	623	0.33*	0.24*	0.15	−0.32*	−0.76*	−1.14*
				[0.20, 0.44]	[0.08, 0.40]	[−0.02, 0.32]	[−0.47, −0.16]	[−1.07, −0.42]	[−1.73, −0.58]

r*-gap only. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A5: Robustness 1d: sample ends in December 2019, start 1996

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	288	−0.06	−0.12	−0.14	−0.03	0.01	0.03
				[−0.29, 0.14]	[−0.46, 0.21]	[−0.53, 0.28]	[−0.12, 0.06]	[−0.16, 0.16]	[−0.19, 0.23]
		6	288	−0.09	−0.17	−0.19	0.00	0.01	0.02
				[−0.41, 0.18]	[−0.70, 0.30]	[−0.78, 0.40]	[−0.11, 0.11]	[−0.19, 0.23]	[−0.25, 0.32]
		12	288	−0.15	−0.24	−0.26	0.08	0.06	0.03
				[−0.34, 0.15]	[−0.61, 0.18]	[−0.77, 0.25]	[−0.05, 0.20]	[−0.11, 0.25]	[−0.20, 0.27]
	r*-gap	3	288	0.24	0.19	0.11	−0.11	−0.10	−0.08
				[−0.02, 0.50]	[−0.18, 0.52]	[−0.28, 0.46]	[−0.26, 0.01]	[−0.31, 0.10]	[−0.35, 0.17]
		6	288	0.23*	0.10	0.02	−0.12	−0.10	−0.10
				[0.00, 0.47]	[−0.28, 0.48]	[−0.42, 0.52]	[−0.25, 0.03]	[−0.30, 0.15]	[−0.37, 0.19]
		12	288	0.30*	0.13	0.00	−0.08	−0.14	−0.21
				[0.10, 0.48]	[−0.15, 0.39]	[−0.36, 0.36]	[−0.23, 0.08]	[−0.42, 0.10]	[−0.60, 0.11]

r*-gap only. All equations, monthly and daily. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A6: Robustness 1e: sample ends in December 2019, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	561	0.03	0.05	0.06	−0.04	−0.06	−0.07
				[−0.06, 0.13]	[−0.08, 0.19]	[−0.09, 0.22]	[−0.18, 0.11]	[−0.43, 0.34]	[−0.67, 0.54]
		6	558	0.11*	0.09	0.04	−0.10	−0.26	−0.44
				[0.01, 0.23]	[−0.05, 0.26]	[−0.11, 0.22]	[−0.23, 0.05]	[−0.56, 0.07]	[−1.06, 0.10]
		12	552	0.19*	0.18*	0.08	−0.21*	−0.49*	−0.82*
				[0.06, 0.34]	[0.02, 0.40]	[−0.13, 0.36]	[−0.34, −0.06]	[−0.82, −0.15]	[−1.39, −0.23]
	r*-gap	3	561	0.04	−0.02	−0.04	−0.02	−0.13	−0.32
				[−0.04, 0.13]	[−0.17, 0.12]	[−0.23, 0.13]	[−0.13, 0.11]	[−0.35, 0.12]	[−0.67, 0.11]
		6	558	0.21*	0.04	−0.02	−0.12	−0.32	−0.56*
				[0.10, 0.32]	[−0.11, 0.21]	[−0.20, 0.17]	[−0.26, 0.02]	[−0.61, 0.01]	[−1.06, −0.07]
		12	552	0.27*	0.16*	0.09	−0.22*	−0.50*	−0.84*
				[0.15, 0.40]	[0.02, 0.36]	[−0.10, 0.33]	[−0.39, −0.07]	[−0.90, −0.21]	[−1.50, −0.38]

r*-gap only. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A7: Robustness 2a: Kiley r*, start 1996

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap (Kiley)	none	3	369	−0.33	−0.35	−0.31	−0.06	−0.04	−0.03
				[−0.59, 0.02]	[−0.63, 0.00]	[−0.62, 0.01]	[−0.27, 0.12]	[−0.35, 0.25]	[−0.41, 0.34]
		6	369	−0.28	−0.28	−0.23	−0.07	−0.12	−0.19
				[−0.54, 0.04]	[−0.63, 0.08]	[−0.64, 0.12]	[−0.27, 0.17]	[−0.54, 0.32]	[−0.78, 0.41]
		12	369	−0.53*	−0.32	−0.23	−0.09	−0.10	−0.13
				[−0.93, −0.14]	[−0.64, 0.00]	[−0.59, 0.17]	[−0.27, 0.08]	[−0.46, 0.20]	[−0.73, 0.31]
	r*-gap (Kiley)	3	369	0.15	−0.06	−0.14	−0.10	−0.07	−0.07
				[−0.16, 0.50]	[−0.29, 0.19]	[−0.30, 0.04]	[−0.34, 0.12]	[−0.44, 0.28]	[−0.50, 0.35]
		6	369	0.34	0.15	0.00	−0.21	−0.27	−0.33
				[−0.08, 0.61]	[−0.20, 0.44]	[−0.30, 0.31]	[−0.45, 0.06]	[−0.73, 0.23]	[−1.02, 0.42]
		12	369	0.39	0.28	−0.01	−0.25*	−0.18	−0.03
				[−0.02, 0.77]	[−0.07, 0.80]	[−0.38, 0.46]	[−0.48, −0.03]	[−0.72, 0.36]	[−0.93, 0.77]

2-year minus expected inflation minus Kiley's r*; the FCI is Note A's index built with Kiley's r* (same adjustment in rate and FCI). Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A8: Robustness 2b: Kiley r*, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap (Kiley)	none	3	642	−0.02	0.00	0.00	−0.04	−0.02	0.01
				[−0.15, 0.08]	[−0.16, 0.18]	[−0.14, 0.19]	[−0.17, 0.10]	[−0.33, 0.29]	[−0.47, 0.51]
		6	639	−0.04	−0.01	−0.01	−0.07	−0.17	−0.28
				[−0.15, 0.06]	[−0.16, 0.16]	[−0.16, 0.16]	[−0.18, 0.04]	[−0.41, 0.08]	[−0.71, 0.15]
		12	633	−0.11	−0.05	−0.03	−0.18*	−0.35*	−0.50*
				[−0.25, 0.03]	[−0.21, 0.12]	[−0.19, 0.14]	[−0.30, −0.06]	[−0.62, −0.11]	[−0.91, −0.11]
	r*-gap (Kiley)	3	642	−0.07	−0.11	−0.11	−0.05	−0.11	−0.18
				[−0.16, 0.06]	[−0.23, 0.02]	[−0.24, 0.01]	[−0.17, 0.06]	[−0.34, 0.09]	[−0.51, 0.12]
		6	639	−0.06	−0.09	−0.07	−0.08	−0.18	−0.30
				[−0.18, 0.06]	[−0.22, 0.04]	[−0.21, 0.08]	[−0.20, 0.04]	[−0.42, 0.07]	[−0.70, 0.07]
		12	633	0.00	−0.04	−0.03	−0.20*	−0.37*	−0.50*
				[−0.14, 0.19]	[−0.17, 0.07]	[−0.16, 0.09]	[−0.33, −0.06]	[−0.63, −0.12]	[−0.88, −0.16]

As 2a. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A9: Robustness 2c: real 2-year and real FCI with a cubic spline trend, start 1996

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Real	none	3	369	−0.44*	−0.40*	−0.34*	−0.07	−0.14	−0.21
				[−0.66, −0.14]	[−0.66, −0.11]	[−0.62, −0.07]	[−0.20, 0.09]	[−0.38, 0.15]	[−0.53, 0.17]
		6	369	−0.44*	−0.34*	−0.24	−0.03	−0.21	−0.41
				[−0.69, −0.13]	[−0.69, −0.02]	[−0.59, 0.10]	[−0.23, 0.15]	[−0.59, 0.15]	[−0.98, 0.11]
		12	369	−0.64*	−0.34	−0.20	0.03	−0.06	−0.21
				[−1.07, −0.26]	[−0.75, 0.06]	[−0.61, 0.28]	[−0.18, 0.20]	[−0.39, 0.29]	[−0.77, 0.29]
	real	3	369	0.00	−0.03	−0.05	−0.17	−0.29	−0.36
				[−0.31, 0.35]	[−0.33, 0.27]	[−0.34, 0.10]	[−0.44, 0.01]	[−0.76, 0.06]	[−0.90, 0.08]
		6	369	0.11	0.20	0.15	−0.20	−0.49*	−0.68*
				[−0.20, 0.57]	[−0.13, 0.65]	[−0.13, 0.45]	[−0.41, 0.01]	[−0.93, −0.03]	[−1.29, −0.01]
		12	369	0.19	0.28	0.17	−0.14	−0.25	−0.27
				[−0.27, 0.63]	[−0.18, 0.78]	[−0.21, 0.62]	[−0.42, 0.08]	[−0.87, 0.21]	[−1.23, 0.39]

No r* measure: the trend is a natural cubic spline in time (knots at the sample ends and at the NBER business-cycle peaks inside the sample), estimated in every monthly equation. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A10: Robustness 2d: real 2-year and real FCI with a cubic spline trend, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Real	none	3	642	0.01	0.04	0.04	−0.11*	−0.22*	−0.29*
				[−0.17, 0.18]	[−0.18, 0.24]	[−0.15, 0.20]	[−0.23, −0.01]	[−0.43, −0.03]	[−0.60, −0.05]
		6	639	0.01	0.06	0.06	−0.11*	−0.31*	−0.47*
				[−0.13, 0.13]	[−0.14, 0.23]	[−0.13, 0.21]	[−0.22, −0.01]	[−0.54, −0.04]	[−0.80, −0.12]
		12	633	−0.10	0.08	0.11	−0.09	−0.25*	−0.38*
				[−0.29, 0.06]	[−0.22, 0.29]	[−0.18, 0.34]	[−0.18, 0.01]	[−0.42, −0.02]	[−0.64, −0.06]
	real	3	642	0.22*	0.15	0.06	−0.17*	−0.31*	−0.38*
				[0.03, 0.37]	[−0.01, 0.28]	[−0.04, 0.14]	[−0.27, −0.04]	[−0.49, −0.06]	[−0.61, −0.07]
		6	639	0.23*	0.21	0.11	−0.17*	−0.43*	−0.61*
				[0.04, 0.42]	[0.00, 0.40]	[−0.01, 0.26]	[−0.31, −0.04]	[−0.69, −0.14]	[−0.95, −0.21]
		12	633	0.18	0.18	0.11	−0.17*	−0.39*	−0.53*
				[0.00, 0.37]	[−0.02, 0.39]	[−0.04, 0.27]	[−0.29, −0.03]	[−0.62, −0.15]	[−0.85, −0.15]

As 2c; knots at the NBER peaks of 1973, 1980, 1981, 1990, 2001, 2007 and 2020. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A11: Robustness 3a: excess bond premium as a third monthly variable, start 1996

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	369	−0.11	−0.14	−0.14	−0.14	−0.17	−0.19
				[−0.43, 0.28]	[−0.54, 0.26]	[−0.54, 0.22]	[−0.35, 0.03]	[−0.55, 0.12]	[−0.69, 0.24]
		6	369	0.02	−0.05	−0.09	−0.19	−0.36	−0.53
				[−0.32, 0.36]	[−0.46, 0.35]	[−0.53, 0.33]	[−0.38, 0.05]	[−0.75, 0.07]	[−1.11, 0.11]
		12	369	−0.08	0.08	0.16	−0.27*	−0.53*	−0.78*
				[−0.42, 0.27]	[−0.38, 0.44]	[−0.43, 0.62]	[−0.41, −0.06]	[−0.88, −0.09]	[−1.38, −0.15]
	r*-gap	3	369	0.19	0.03	−0.14	−0.18	−0.20	−0.19
				[−0.14, 0.51]	[−0.40, 0.43]	[−0.63, 0.27]	[−0.38, 0.03]	[−0.62, 0.21]	[−0.76, 0.36]
		6	369	0.31	0.22	0.04	−0.29*	−0.55*	−0.76*
				[−0.05, 0.65]	[−0.15, 0.67]	[−0.38, 0.53]	[−0.53, −0.08]	[−0.99, −0.13]	[−1.49, −0.09]
		12	369	0.27	0.52*	0.42	−0.39*	−0.78*	−1.10*
				[−0.24, 0.79]	[0.01, 0.93]	[−0.08, 0.86]	[−0.59, −0.16]	[−1.21, −0.32]	[−1.81, −0.30]

r*-gap. Monthly system u, inflation, EBP (EBP treated like u and inflation; responses shown for u and the price level). Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A12: Robustness 3b: excess bond premium as a third monthly variable, start 1973

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
r*-gap	none	3	642	0.00	0.04	0.04	−0.09	−0.15	−0.19
				[−0.09, 0.15]	[−0.11, 0.22]	[−0.09, 0.20]	[−0.24, 0.05]	[−0.52, 0.23]	[−0.73, 0.42]
		6	639	0.01	0.03	0.01	−0.12*	−0.31*	−0.51*
				[−0.11, 0.13]	[−0.14, 0.19]	[−0.19, 0.16]	[−0.26, −0.02]	[−0.60, −0.04]	[−1.07, −0.09]
		12	633	−0.03	−0.01	0.01	−0.22*	−0.50*	−0.78*
				[−0.17, 0.14]	[−0.18, 0.16]	[−0.19, 0.18]	[−0.38, −0.11]	[−0.83, −0.23]	[−1.34, −0.29]
	r*-gap	3	642	0.03	0.01	−0.02	−0.10	−0.20	−0.33
				[−0.10, 0.13]	[−0.15, 0.15]	[−0.18, 0.14]	[−0.23, 0.08]	[−0.54, 0.17]	[−0.82, 0.26]
		6	639	0.04	0.02	0.00	−0.12	−0.32	−0.58
				[−0.10, 0.18]	[−0.17, 0.18]	[−0.22, 0.15]	[−0.27, 0.03]	[−0.64, 0.02]	[−1.09, 0.03]
		12	633	0.06	0.03	0.04	−0.24*	−0.57*	−0.89*
				[−0.12, 0.24]	[−0.19, 0.23]	[−0.17, 0.25]	[−0.40, −0.08]	[−0.92, −0.22]	[−1.43, −0.39]

As 3a. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A13: Robustness 4: the FCI replaced by its three weighted components

				Unemployment gap (pp)			Price level (%)		
Case	FCI	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	components	3	369	−0.08	−0.04	−0.04	0.07	0.11	0.09
				[−0.41, 0.28]	[−0.35, 0.23]	[−0.25, 0.12]	[−0.17, 0.31]	[−0.29, 0.55]	[−0.44, 0.69]
		6	369	0.08	0.23	0.12	−0.11	−0.20	−0.26
				[−0.36, 0.46]	[−0.25, 0.61]	[−0.18, 0.39]	[−0.41, 0.19]	[−0.81, 0.44]	[−1.08, 0.66]
		12	369	0.03	0.25	0.09	0.03	0.21	0.42
				[−0.49, 0.56]	[−0.21, 0.70]	[−0.31, 0.41]	[−0.25, 0.29]	[−0.36, 0.68]	[−0.54, 1.17]
Real	components	3	369	0.09	−0.01	−0.06	−0.03	0.00	−0.02
				[−0.23, 0.39]	[−0.28, 0.17]	[−0.24, 0.04]	[−0.21, 0.20]	[−0.28, 0.38]	[−0.36, 0.47]
		6	369	0.25	0.21	0.01	−0.11	−0.14	−0.14
				[−0.05, 0.63]	[−0.05, 0.52]	[−0.17, 0.20]	[−0.39, 0.14]	[−0.66, 0.34]	[−0.81, 0.50]
		12	369	0.17	0.29	−0.03	−0.06	0.14	0.41
				[−0.32, 0.63]	[−0.11, 0.65]	[−0.33, 0.23]	[−0.29, 0.18]	[−0.36, 0.61]	[−0.32, 1.08]
r*-gap	components	3	369	0.05	−0.10	−0.16*	0.06	0.23	0.29
				[−0.19, 0.40]	[−0.33, 0.18]	[−0.35, −0.01]	[−0.17, 0.26]	[−0.13, 0.61]	[−0.15, 0.82]
		6	369	0.23	0.12	−0.05	−0.13	−0.10	−0.07
				[−0.03, 0.64]	[−0.15, 0.49]	[−0.31, 0.20]	[−0.36, 0.11]	[−0.54, 0.42]	[−0.81, 0.59]
		12	369	0.20	0.22	−0.09	−0.14	0.06	0.35
				[−0.20, 0.69]	[−0.14, 0.61]	[−0.42, 0.21]	[−0.36, 0.04]	[−0.40, 0.47]	[−0.40, 0.98]

The 10-year yield, equity valuation and the real dollar, each multiplied by its weight, enter the daily and monthly equations separately, in the same step as the 2-year yield; each has its own window impact. Replaces the FCI, so the same step without the FCI is the baseline row. Per +1 pp on the 2-year yield. * = 90% interval excludes zero; intervals in brackets. Obs = monthly observations in the equations.

Table A16: Monthly VAR benchmark (3, 6 and 12 lags)

					Unemployment gap (pp)			Price level (%)		
2-year	FCI	Identification	Lags	Obs	12 m	24 m	36 m	12 m	24 m	36 m
Nominal	none	Proxy VAR	3	369	−0.53	−0.43*	−0.35*	0.00	−0.03	−0.06
					[−0.90, 0.06]	[−0.64, −0.10]	[−0.49, −0.13]	[−1.08, 0.77]	[−1.14, 0.76]	[−1.16, 0.72]
			6	369	−0.46	−0.36	−0.26	−0.05	−0.15	−0.27
					[−0.93, 0.36]	[−0.76, 0.34]	[−0.57, 0.27]	[−1.54, 0.85]	[−2.09, 1.02]	[−2.34, 0.98]
			12	369	−0.91*	−0.61*	−0.32	0.03	−0.08	−0.26
					[−1.37, −0.42]	[−1.08, −0.05]	[−0.71, 0.17]	[−1.20, 0.97]	[−1.83, 1.25]	[−2.27, 1.25]
		Window impact	3	369	−0.61*	−0.46*	−0.36	0.14	0.12	0.09
					[−0.87, −0.26]	[−0.87, −0.06]	[−0.85, 0.01]	[−0.04, 0.34]	[−0.15, 0.50]	[−0.26, 0.61]
			6	369	−0.54*	−0.42*	−0.30	0.09	0.03	−0.07
					[−0.89, −0.27]	[−0.81, −0.14]	[−0.77, 0.00]	[−0.09, 0.31]	[−0.26, 0.37]	[−0.54, 0.43]
			12	369	−0.90*	−0.63*	−0.35	0.13	0.06	−0.09
					[−1.45, −0.48]	[−1.17, −0.26]	[−0.95, 0.07]	[−0.10, 0.39]	[−0.41, 0.56]	[−0.78, 0.66]
	nominal	Proxy VAR	3	369	0.35	0.13	0.02	−0.29	−0.42	−0.48
					[−0.34, 173.16]	[−0.40, 100.92]	[−0.38, 62.73]	[−73.88, 0.60]	[−87.91, 0.46]	[−90.42, 0.40]
			6	369	0.63	0.37	0.17	−0.49	−0.78	−0.95
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
			12	369	0.72	0.51	0.32	−0.67	−1.02	−1.17
					unbounded	unbounded	unbounded	unbounded	unbounded	unbounded
		Window impact	3	369	−0.06	0.02	0.00	−0.06	−0.22	−0.32
					[−0.34, 0.23]	[−0.33, 0.38]	[−0.37, 0.46]	[−0.24, 0.12]	[−0.56, 0.11]	[−0.85, 0.13]
			6	369	0.02	0.08	0.05	−0.12	−0.34	−0.52
					[−0.32, 0.35]	[−0.29, 0.60]	[−0.33, 0.63]	[−0.37, 0.11]	[−0.81, 0.09]	[−1.23, 0.06]
			12	369	−0.14	−0.09	−0.01	−0.13	−0.36	−0.57
					[−0.57, 0.37]	[−0.69, 0.45]	[−0.64, 0.47]	[−0.39, 0.08]	[−0.80, 0.07]	[−1.23, 0.06]
Real	none	Proxy VAR	3	369	−0.45*	−0.41*	−0.35*	−0.04	−0.06	−0.08
					[−0.88, −0.13]	[−0.65, −0.22]	[−0.52, −0.22]	[−0.68, 0.62]	[−0.72, 0.62]	[−0.73, 0.59]
			6	369	−0.48*	−0.38	−0.26	−0.06	−0.17	−0.31
					[−1.04, −0.05]	[−0.84, 0.03]	[−0.60, 0.06]	[−0.91, 0.76]	[−1.29, 0.89]	[−1.51, 0.82]
			12	369	−0.72*	−0.48*	−0.22	−0.12	−0.29	−0.46
					[−1.29, −0.35]	[−0.99, −0.07]	[−0.62, 0.15]	[−0.97, 0.71]	[−1.50, 0.85]	[−1.85, 0.80]
		Window impact	3	369	−0.38*	−0.37*	−0.32*	−0.04	−0.06	−0.07
					[−0.63, −0.09]	[−0.66, −0.05]	[−0.68, −0.03]	[−0.23, 0.14]	[−0.37, 0.24]	[−0.45, 0.33]
			6	369	−0.41*	−0.34	−0.24	−0.03	−0.13	−0.26
					[−0.77, −0.09]	[−0.77, 0.00]	[−0.74, 0.13]	[−0.28, 0.27]	[−0.54, 0.43]	[−0.82, 0.53]
			12	369	−0.65*	−0.46*	−0.23	−0.04	−0.16	−0.29
					[−1.18, −0.23]	[−1.01, −0.02]	[−0.77, 0.30]	[−0.27, 0.18]	[−0.62, 0.26]	[−1.01, 0.36]
	real	Proxy VAR	3	369	0.26	0.01	−0.09	−0.23	−0.30	−0.32
					[−0.27, 2.00]	[−0.40, 1.23]	[−0.38, 0.77]	[−1.28, 0.48]	[−1.48, 0.40]	[−1.54, 0.37]
			6	369	0.49	0.26	0.07	−0.46	−0.72	−0.85
					[−0.22, 4.80]	[−0.35, 3.58]	[−0.34, 2.24]	[−3.03, 0.44]	[−4.25, 0.40]	[−4.56, 0.31]
			12	369	0.72	0.49	0.30	−0.72	−1.05	−1.12
					[−0.21, 24.27]	[−0.33, 19.69]	[−0.25, 12.47]	[−11.42, 0.16]	[−16.08, 0.22]	[−16.57, 0.33]
		Window impact	3	369	0.04	−0.04	−0.10	−0.18	−0.28	−0.32
					[−0.27, 0.40]	[−0.38, 0.41]	[−0.48, 0.31]	[−0.40, 0.03]	[−0.66, 0.05]	[−0.80, 0.12]
			6	369	0.06	0.04	−0.03	−0.23	−0.45	−0.60*
					[−0.28, 0.42]	[−0.38, 0.42]	[−0.47, 0.41]	[−0.47, 0.04]	[−0.94, 0.01]	[−1.23, −0.04]
			12	369	−0.02	−0.04	0.00	−0.26*	−0.49	−0.63
					[−0.51, 0.47]	[−0.61, 0.46]	[−0.60, 0.58]	[−0.63, −0.01]	[−1.11, 0.02]	[−1.53, 0.10]
r*-gap	none	Proxy VAR	3	369	−0.27	−0.22*	−0.17*	−0.10	−0.13	−0.15
					[−0.80, 0.06]	[−0.51, −0.03]	[−0.34, −0.06]	[−0.71, 0.59]	[−0.75, 0.57]	[−0.77, 0.55]
			6	369	−0.21	−0.11	−0.03	−0.18	−0.38	−0.56
					[−0.86, 0.18]	[−0.61, 0.22]	[−0.36, 0.21]	[−0.88, 0.67]	[−1.28, 0.68]	[−1.49, 0.54]
			12	369	−0.33	−0.10	0.06	−0.29	−0.57	−0.79
					[−0.98, 0.02]	[−0.64, 0.24]	[−0.30, 0.32]	[−0.97, 0.48]	[−1.51, 0.47]	[−1.83, 0.31]
		Window impact	3	369	−0.16	−0.17	−0.14	−0.13	−0.15	−0.17
					[−0.53, 0.27]	[−0.66, 0.44]	[−0.67, 0.42]	[−0.33, 0.08]	[−0.56, 0.21]	[−0.71, 0.32]
			6	369	−0.06	−0.02	0.03	−0.24	−0.45	−0.61
					[−0.54, 0.38]	[−0.55, 0.53]	[−0.50, 0.61]	[−0.48, 0.00]	[−1.04, 0.05]	[−1.44, 0.02]
			12	369	−0.19	−0.03	0.09	−0.27*	−0.51*	−0.69*
					[−0.70, 0.31]	[−0.53, 0.43]	[−0.27, 0.61]	[−0.51, −0.05]	[−0.99, −0.08]	[−1.38, −0.07]
	r*-gap	Proxy VAR	3	369	0.33	0.13	−0.01	−0.19	−0.24	−0.25
					[−0.15, 1.34]	[−0.26, 0.93]	[−0.30, 0.59]	[−0.89, 0.48]	[−0.95, 0.43]	[−0.98, 0.40]
			6	369	0.67	0.41	0.16	−0.59	−0.87	−1.00*
					[0.00, 2.51]	[−0.12, 1.84]	[−0.20, 1.11]	[−1.80, 0.12]	[−2.40, 0.02]	[−2.53, −0.09]
			12	369	0.95*	0.73	0.47	−0.84*	−1.24*	−1.35*
					[0.01, 5.87]	[−0.04, 4.55]	[−0.05, 2.89]	[−3.17, −0.15]	[−4.39, −0.24]	[−4.51, −0.24]
		Window impact	3	369	0.19	0.06	−0.05	−0.19	−0.24	−0.26
					[−0.16, 0.56]	[−0.42, 0.43]	[−0.53, 0.32]	[−0.42, 0.06]	[−0.66, 0.22]	[−0.80, 0.37]
			6	369	0.36	0.22	0.04	−0.41*	−0.67*	−0.81*

the middles of adjacent months, and add it to the daily yield. The daily index is

$$F^v_\tau = w_B\, y^{10}_\tau + w_Q\, 100 \ln \mathrm{CAPE}_\tau + w_D\, 100 \ln \mathrm{DOL}_\tau + o^v_\tau, \tag{18}$$

where DOL_τ is the daily nominal broad dollar. The offset o^v_τ is the monthly difference between the index in equation (4) and the average of the first three terms, interpolated in the same way. It also absorbs the difference between the real and the nominal dollar. The monthly averages of the daily series are very close to the monthly measures: the root-mean-square difference is at most 0.02 unit for the FCI and 0.04 percentage point for the two-year yield. The offsets change little from day to day, so day-to-day moves in the index come from market prices. Table A17 shows the measures used at each frequency. The daily calendar has every weekday; missing values, such as on holidays, are filled from the previous observation for up to three days. Each daily equation has 13 coefficients with the FCI; each monthly equation has $1 + 4L$ with the FCI and $1 + 3L$ without it.

Table A17: Measures of stocks and the dollar at each frequency

Frequency	Stocks	Dollar
Window	S&P 500	DXY index
Daily	CAPE	Nominal broad dollar, plus offset
Monthly	CAPE, or earnings-yield gap to r^*	Real broad dollar

Tracing the effect of a surprise. The response is computed recursively, as a deviation from a path with no surprise. Let the announcement occur on weekday d_0 of the announcement month, and let $h = 0, 1, \ldots, 36$ count months from then. Tildes denote responses.

1. In each month, unemployment and inflation come first: $\tilde{m}_0 = 0$, and for $h \geq 1$,

$$\tilde{m}_h = \sum_{l=1}^{L} A_l\, \tilde{m}_{h-l} + \sum_{l=1}^{L} G_l\, \tilde{x}^M_{h-l}, \tag{19}$$

   with terms dated before the announcement month equal to zero, and $\tilde{x}^M_h$ the response of the monthly average of x.

2. The days of the month follow. Before day d_0, $\tilde{x}_\tau = 0$. On day d_0, $\tilde{x}_\tau = b$ from equation (13). On every later day, with $\tilde{x}^{(k)}_{\tau-1}$ the average of the responses over the k days through $\tau - 1$,

$$\tilde{x}_\tau = \sum_{k \in \{1,5,22,66\}} \Phi_k\, \tilde{x}^{(k)}_{\tau-1} + \Gamma_0\, \tilde{m}_{h(\tau)} + \Gamma_1\, \tilde{m}_{h(\tau)-1}, \tag{20}$$

   where $h(\tau)$ is the month of day τ, counted from the announcement month.

3. The monthly average $\tilde{x}^M_h$ then enters equation (19) for later months.

Months after the announcement month have 21 weekdays. To avoid computing a path for every date, we form six groups of announcement dates by the weekday of the month, in blocks of four days, and compute the path at each group's average day and month length. With ω_g the share of announcements in group g, the reported response is $\sum_g \omega_g\, \tilde{m}^{(g)}_h$. Because the model is linear, these responses do not depend on the constant or the trend.

Bootstrap. In each of 200 draws, we resample the 252 announcements with replacement, which gives new values b^* in equation (13) and new group shares ω^*_g. We also resample the residuals $\hat{\zeta}_t$ of equation (15) jointly across equations, add them to the fitted values, holding the regressors

fixed, and re-estimate the monthly block. The daily block is held at its estimate. We compute the response for each draw and report percentile intervals. The sets for the Phillips multiplier in Table 4 use the same 200 draws, with the announcement draws shared across steps.

C Data sources

Table A18 lists the data. Series from FRED are identified by their mnemonics. Quarterly series are placed in the middle month of the quarter and interpolated linearly.

Table A18: Data sources

Series	Frequency	Source
Unemployment rate	Monthly	FRED UNRATE
Industrial production; commodity prices	Monthly	FRED INDPRO; IMF, FRED PALLFNFINDEXM
VIX	Daily	FRED VIXCLS
Natural rate of unemployment	Quarterly	CBO, FRED NROU
Core PCE price index	Monthly	FRED PCEPILFE
Two-year Treasury yield	Daily	FRED DGS2
10-year Treasury yield	Daily, monthly	FRED DGS10, GS10
Expected inflation, 2 and 10 years	Monthly	Cleveland Fed, FRED EXPINF2YR, EXPINF10YR
r^*, Laubach–Williams	Quarterly	New York Fed, one-sided estimates
CAPE	Monthly	Shiller, spliced over 1990–92 to the monthly average of daily CAPE
Daily CAPE	Daily	S&P 500 (FRED SP500; before October 2016, Dow Jones S&P 500 history, level-matched) over Shiller's 10-year earnings
Real broad dollar	Monthly	FRED RTWEXBGS, spliced to TWEXBPA in 2006
Nominal broad dollar	Daily	FRED DTWEXBGS, earlier DTWEXB, TWEXBMTH
Futures, yields, S&P 500 and DXY in the window	Event	U.S. Monetary Policy Event-Study Database, "monetary events" (Acosta et al., 2025)